

COMPUTING HOMOTOPY GROUPS OF MADSEN-TILLMANN SPECTRA

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1. INTRODUCTION

In [GMTW09], Galatius, Madsen, Tillmann, and Weiss defined for each nonnegative integer $d \geq 0$, a spectrum $MT\theta(d)$, whose infinite loop space deloops to the classifying space $BCob(d)_\theta$ of a particular topological category. The category $Cob(d)_\theta$ parametrizes cobordisms between $(d-1)$ -dimensional closed smooth manifolds with θ -structures for a given tangential structure $\theta : B \rightarrow BO(d)$. The homotopy invariants of these spectra, called *Madsen-Tillmann spectra*, contain information about diffeomorphism groups of manifolds [GRW10], as well as applications to physics via topological quantum field theories [FH21]. These applications have sparked great interest in using techniques from stable homotopy theory to better understand and classify these spectra.

One of the most effective means of studying Madsen-Tillman spectra came in the work of Kashiwabara and Zare [KZ18], which made use of the transfer maps of [BG74] and [MMM86] to give splittings of several classes of Madsen-Tillmann spectra localized at various primes. We focus our attention on the unoriented Madsen-Tillmann spectra $MTO(n)$. Kashiwabara and Zare determined the homotopy type of $MTO(n)$ away from the prime 2.

Theorem 1.0.1. *Given $n \geq 0$, we have a homotopy equivalence*

$$MTO(n) \left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix} \right] \simeq \begin{cases} \Sigma_+^\infty BO(2n) \left[\begin{smallmatrix} 1 \\ 2 \end{smallmatrix} \right], & n \text{ even}, \\ *, & n \text{ odd}. \end{cases}$$

In the case $n = 2$, in a follow-up paper [KZ15], Kashiwabara and Zare were able to completely determine the homotopy type of $MTO(2)$ completed at the prime 2, hence giving a complete determination of its homotopy type.

Theorem 1.0.2. [KZ15] *We have a homotopy equivalence of spectra*

$$MTO(2)_2^\wedge \simeq \Sigma_+^\infty BSO(3)_2^\wedge \vee \Sigma^{-2} D(2).$$

We use the above splitting theorem to extend the range of known computations of the 2-primary homotopy groups of $MTO(2)$ (modulo some extension problems) through dimensions 15.

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2. A BRIEF RECOLLECTION OF HOMOLOGICAL STABILITY AND MAPPING CLASS GROUPS

2.1. The oriented case.

The motivation behind the definition of Madsen-Tillmann spectra originates in the study of the (stable) mapping class group. We begin by fixing some notation.

Notation 2.1.1.

1. Let $\Sigma_{g,b}$ denote a connected compact oriented surface of genus g with b many boundary components.
2. Given an orientable manifold M possibly with boundary, let $\text{Diff}_\partial^+(M)$ denote the group of orientation-preserving diffeomorphisms of M which fix the boundary ∂M pointwise.

Definition 2.1.2. The *mapping class group* $\Gamma_{g,b}$ is the group of isotopy classes of oriented diffeomorphisms of $\Sigma_{g,b}$ which fix the boundary pointwise:

$$\Gamma_{g,b} := \pi_0(\text{Diff}_\partial^+(\Sigma_{g,b})).$$

We write Γ_g for $\Gamma_{g,0}$.

Note since the diffeomorphism type of a smooth surface is determined by its homeomorphism type, the pair (g,b) defines $\Sigma_{g,b}$ up to diffeomorphism, and hence implies $\Gamma_{g,b}$ is well defined up to isomorphism.

One motivation behind studying such groups is the following. Given $g > 1$, there is a contractible space $\mathcal{H}_{g,b}$ of certain hyperbolic metrics¹ on $\Sigma_{g,b}$, which admits an action by $\text{Diff}_\partial^+(\Sigma_{g,b})$. Given such metric $\omega : T\Sigma_{g,b}^{\oplus 2} \rightarrow \mathbf{R}$, $f \in \text{Diff}_\partial^+(\Sigma_{g,b})$ acts via pullback along the differential:

$$f \cdot \omega := (df \oplus df)^* \omega.$$

¹By definition, these are Riemannian metrics of constant sectional curvature -1.

The orbit space is the *moduli space of Riemannian surfaces with topological type* $\Sigma_{g,b}$:

$$\mathcal{M}_{g,b} = \mathcal{H}_{g,b} / \text{Diff}_{\partial}^+(\Sigma_{g,b}).$$

For $b > 0$, this action is free, hence $\mathcal{M}_{g,b}$ is the classifying space for principal $\text{Diff}_{\partial}^+(\Sigma_{g,b})$ -bundles. We can get a free action for all $b \geq 0$ by restricting to the basepoint component $\text{Diff}_{\partial,1}^+(\Sigma_{g,b})$, which yields a principal $\text{Diff}_{\partial,1}^+(\Sigma_{g,b})$ -bundle over the base space

$$\mathcal{T}_{a,b} := \mathcal{H}_{a,b} / \text{Diff}_{\partial,1}^+(\Sigma_{g,b}).$$

This space, parametrizing hyperbolic metrics on $\Sigma_{g,b}$ up to pullback along diffeomorphisms isotopic to the identity, is the well-studied *Teichmüller space of topological type* $\Sigma_{g,b}$. In particular, $\mathcal{T}_{g,b}$ is known to be homeomorphic to the Euclidean space $\mathbf{R}^{6g-6+2b}$, and so by the long exact sequence on homotopy groups associated to the above fiber bundle, we conclude $\text{Diff}_{\partial,1}^+(\Sigma_{g,b})$ is contractible.

Viewing π_0 as a functor valued in discrete spaces, for $b > 0$, projection onto path components defines a homotopy equivalence $\Gamma_{g,b} \simeq \text{Diff}_{\partial}^+(\Sigma_{g,b})$. Thus, $B\Gamma_{g,b}$ is also a model for the classifying space for smooth surface bundles with fiber $\Sigma_{g,b}$, and so has the homotopy type of $\mathcal{M}_{g,b}$.

There are various maps between the oriented surfaces $\Sigma_{g,b}$ for different values of (g,b) , which induce associated homomorphisms between their mapping class groups. Namely, for each pair (g,b) with $b > 0$ for the first two constructions, and $b > 1$ for the latter two, we have

- a map $a: \Sigma_{g,b} \rightarrow \Sigma_{g,b+1}$ given by gluing (in an orientation-preserving manner) a pair of pants (i.e., a $\Sigma_{0,3}$) to $\Sigma_{g,b}$ along a common boundary component;
- a map $b: \Sigma_{g,b} \rightarrow \Sigma_{g,b-1}$ given by gluing a disc (i.e., a $\Sigma_{0,1}$) along a common boundary component;
- a map $d: \Sigma_{g,b} \rightarrow \Sigma_{g+1,b-1}$ given by gluing a pair of pants to $\Sigma_{g,b}$ along two common boundary components;
- a map $g: \Sigma_{g,b} \rightarrow \Sigma_{g+1,b-2}$ given by gluing a cylinder (i.e., a $\Sigma_{0,2}$) to $\Sigma_{g,b}$ along two common boundary components.

Such gluing constructions induce inclusions of $\Sigma_{g,b}$ into the various codomains of the above maps. Consequently, we get group homomorphisms

$$\begin{aligned} \alpha: \Gamma_{g,b} &\rightarrow \Gamma_{g,b+1} \\ \beta: \Gamma_{g,b} &\rightarrow \Gamma_{g,b-1} \\ \delta: \Gamma_{g,b} &\rightarrow \Gamma_{g+1,b-1} \\ \gamma: \Gamma_{g,b} &\rightarrow \Gamma_{g+1,b-2} \end{aligned}$$

given by sending $[f] \in \Gamma_{g,b}$ to the isotopy class of the diffeomorphism $\tilde{f}$ defined to be f on $\Gamma_{g,b}$, and the identity map on the complement. The map β is a left inverse of α .

Definition 2.1.3.

Harer's *homological stability theorem* [Ref] states that α, δ , and γ induce isomorphisms on integral homology groups in dimensions $k \leq (g-3)/3$. Miller showed that β induces a homology isomorphism for $k \leq (g-4)/3$. Shortly after, Ivanov extended these computations, improving Harer's bounds to $(g-1)/2$ and $(g-2)/2$ respectively. Given $k \geq 0$, we refer to those values of $g \geq 0$ for which we know the above maps induce isomorphisms on $H_k(-; \mathbf{Z})$ (i.e., $g \geq 2k+2$ by Ivanov's work) as the *stable range*. Thus $H_k(\Gamma_{g,n}; \mathbf{Z})$ is independent of b and g for g within the stable range. These are called the *stable homology groups* of the mapping class groups.

Definition 2.1.4. The *stable mapping class group* $\Gamma_{\infty,b}$ is defined as the colimit

$$\Gamma_{\infty,b} := \text{colim}_{n \rightarrow \infty} \Gamma_{n,b},$$

where the structure homomorphisms $\Gamma_{n,b} \rightarrow \Gamma_{n+1,b}$ are given by the composite

$$\Gamma_{n,b} \xrightarrow{\alpha} \Gamma_{n,b+1} \xrightarrow{\delta} \Gamma_{n+1,b}.$$

Again, for $b = 0$, we suppress the subscript, writing Γ_{∞} .

The homology of the stable mapping class group Γ_∞ is the stable homology of the mapping class groups, i.e. the stability theorem gives isomorphisms

$$H_k(\Gamma_\infty) \cong H_k(\Gamma_{g,b})$$

for $g \geq 2k + 1$ if $b \geq 1$ and $g \geq 2k$

Powell [ref P] showed that Γ_g was a perfect group (i.e., its abelianization G^{ab} is trivial) for $g \geq 3$, and hence Γ_∞ is also perfect (as $(-)^{\text{ab}}$ is a left adjoint and hence commutes with colimits). In particular, the classifying space $B\Gamma_\infty$ is a connected aspherical CW-complex with $\pi_1(B\Gamma_\infty) \cong \pi_0(\Gamma_\infty) = \Gamma_\infty$.

A group G is *hypoabelian* if G contains no nontrivial perfect subgroup. Let $P \leq G$ be the largest perfect subgroup (this necessarily exists as the subgroup generated by a collection of perfect subgroups is itself perfect). Then P is necessarily normal and G/P is hypoabelian. Furthermore the projection $G \rightarrow G/P$ is the universal group homomorphism from G to a hypoabelian group. We call a pair $(H, f: G \rightarrow H)$ satisfying such a universal property the *hypoabelianization* of G . Given a connected CW-complex X , the *Quillen plus construction* gives a procedure to attach 2- and 3-cells to X in order to obtain a new space X^+ together with an acyclic map $f: X \rightarrow X^+$ such that

$$f_*: \pi_1(X) \rightarrow \pi_1(X^+)$$

is the hypoabelianization of $\pi_1(X)$. The pair (X^+, f) is unique up to a unique homotopy equivalence. Applying the Quillen plus construction to the classifying space of the infinite mapping class group yields a simply connected space $B\Gamma_\infty^+$.

Importantly, the effect the Quillen plus construction has on the higher homotopy groups of the initial space is in general not clear. In the case of Γ_∞^+ , at the expense of killing off π_1 , we have created infinitely many nontrivial homotopy groups. An upside to passing to the plus construction is that for $b \geq 1$, the homology equivalence $B\Gamma_{\infty,b} \rightarrow B\Gamma_\infty$ induced by the composite β^b induces homotopy equivalences $B\Gamma_\infty^+ \simeq B\Gamma_{\infty,b}^+$ by Whitehead's theorem, or alternatively, via the universal property of the plus construction.

In [Til97], Tillmann determined $\mathbf{Z} \times B\Gamma_\infty^+$ has the homotopy type of $\Omega B\mathcal{C}$, where $\mathcal{C}$ was a particular symmetric monoidal simplicial cobordism category of oriented surfaces. Segal's infinite loop space machinery implies that the classifying space of any topological symmetric monoidal category $\mathcal{M}$ is an E_∞ -space. In particular, the corresponding loop space $\Omega B\mathcal{M}$ is grouplike, hence Tillmann showed $\mathbf{Z} \times B\Gamma_\infty^+$ is an infinite loop space. One is then naturally led to ask if there exist any interesting spectra whose infinite loop space is $\mathbf{Z} \times B\Gamma_\infty^+$. This is precisely where Madsen-Tillmann spectra (to be defined in section 2) enter the picture.

2.2. The non-orientable case.

There is an analogous mapping class group construction for non-orientable surfaces. Let $N_{g,b}$ denote a connected compact non-orientable surface of (non-orientable) genus g^2 and b boundary components. By the classification of compact connected surfaces, $N_{g,b}$ has the diffeomorphism type of the g -fold connected sum $(\mathbf{RP}^2)^{\#g}$ with b disjoint discs removed. We define the *non-orientable mapping class group* $\mathcal{N}_{g,b}$ of $N_{g,b}$ analogously to the oriented case 2.1.2, though of course we drop the requirement of the diffeomorphisms being orientation-preserving. We suppress the subscript b in the case $b = 1$. Again, for pairs (g, b) with $b > 0$, we have stabilization homomorphisms between the various mapping class groups $\mathcal{N}_{g,b}$:

$$\begin{aligned} \alpha: \mathcal{N}_{g,b} &\rightarrow \mathcal{N}_{g,b+1} \\ \beta: \mathcal{N}_{g,b} &\rightarrow \mathcal{N}_{g,b-1} \\ \gamma: \mathcal{N}_{g,b} &\rightarrow \mathcal{N}_{g+1,b} \end{aligned}$$

The maps α, β, γ are induced respectively by gluing a pair of pants, a disc, or an $N_{1,2}$ to $N_{g,b}$ along a common boundary component. Again, β is a left inverse of α for $b \geq 2$.

Natalie Wahl [Wah08] established an analogue of Harer's homological stability theorem for the non-orientable mapping class groups.

Theorem 2.2.1. *For $b \geq 1$, we have homology isomorphisms*

$$\begin{aligned} \alpha_k: H_k(\mathcal{N}_{g,b}) &\xrightarrow{\cong} H_k(\mathcal{N}_{g,b+1}) \\ \beta_{k'}: H_{k'}(\mathcal{N}_{g,1}) &\xrightarrow{\cong} H_{k'}(\mathcal{N}_{g,0}) \\ \gamma_k: H_k(\mathcal{N}_{g,b}) &\xrightarrow{\cong} H_k(\mathcal{N}_{g+1,b}) \end{aligned}$$

²The non-orientable genus $k(N)$ for a non-orientable surface N is defined via the Euler characteristic as $k(N) := 2 - \chi(N)$.

for $k \leq (g-3)/4$ and $k' \leq (g-5)/4$.

We refer to $g \geq 4k+5$ as the *(non-orientable) stable range*. As in the orientable case, we construct the *non-orientable stable mapping class group* $\mathcal{N}_{\infty,b} := \operatorname{colim}_{g \rightarrow \infty} \mathcal{N}_{g,b}$ ³ to encode the stable homology groups of the non-orientable mapping class groups:

$$H_k(\mathcal{N}_{\infty}) \cong H_k(\mathcal{N}_{g,b})$$

for all pairs g, b with g in the stable range.

Although $\mathcal{N}_{\infty}$ is not a perfect group as in the oriented case, its maximal normal subgroup has index 2, and as before we can apply the Quillen plus construction to yield a space $B\mathcal{N}_{\infty}^+$. This space is also an infinite loop space, corresponding to a spectrum $MTO(2)$ which we define in the next section.

3. TANGENTIAL STRUCTURES AND MADSEN-TILLMANN SPECTRA

3.1. A word on classifying spaces and principal G -bundles.

Throughout the paper, we make use of several standard results concerning principal G -bundles and vector bundles. Those which will be used repeatedly or which we believe may not be familiar to many readers are summarized below. All of the results can be found in [Mit17].

Definition 3.1.1. A subgroup H of a topological group G is *admissible* if the quotient map $G \rightarrow G/H$ is a principal H -bundle.

The main instance of admissible subgroups we will be interested in is the following.

Proposition 3.1.2. A closed subgroup K of a Lie group G is admissible.

One of the main applications of admissible subgroups is the following.

Proposition 3.1.3. Suppose $K \leq G$ is an admissible subgroup. Letting $i: K \hookrightarrow G$ denote the inclusion homomorphism, we have a weak homotopy equivalence $BK \simeq EG/K \cong EG \times_G G/K$, and furthermore under this model, the map $Bi: BK \rightarrow BG$ is given by the projection of the fiber bundle associated to EG with fiber G/K .

The following fact will be used repeatedly throughout our proofs.

Proposition 3.1.4. Let E be a principal G -space over B and F a (left) G -space. Given a continuous map $f: B' \rightarrow B$, we have a natural isomorphism of fiber bundles with structure group G :

$$f^*(E \times_G F) \cong f^*(E) \times_G F.$$

3.1.1. Classifying spaces. Throughout, let $K(n)$ denote either $O(n)$ or $SO(n)$ for some fixed $n \geq 0$. We let $BK(n)$ denote the classifying space for n -dimensional real vector bundles with $K(n)$ -structure. Recall, given a CW-complex B , a real vector bundle V of rank n over B is classified by a map $\xi: B \rightarrow BGL_n(\mathbf{R})$, and a $K(n)$ -structure on V , for $K(n) \subset GL_n(\mathbf{R})$, amounts to a reduction of the structure group of V to $K(n)$, or equivalently, the existence (and choice) of a homotopy-coherent diagram⁴ of the following form:

$$\begin{array}{ccc} & & K(n) \\ & \nearrow \tilde{\xi} & \downarrow \\ B & \xrightarrow{\xi} & BGL_n(\mathbf{R}) \end{array} \quad .$$

$\gamma \uparrow$

The choice of homotopy γ corresponds to a choice of isomorphism of vector bundles

$$\tilde{\xi}^*(\gamma_n^K) \times_{K(n)} \mathbf{R}^n \cong V,$$

where γ_n^K denotes the universal principal $K(n)$ -bundle. The Gram-Schmidt process defines a homotopy equivalence $BGL_n(\mathbf{R}) \simeq BO(n)$, which corresponds to the geometric fact that vector bundles over a paracompact space can be equipped with a Riemannian metric. Thus we can consider rank n vector bundles as being classified by maps into $BO(n)$.

³Identifying $\mathcal{N}_{g,b}$ with $N_{0,b} \# (\mathbf{RP}^2)^{\#g}$, the structure map $\mathcal{N}_{g,b} \rightarrow \mathcal{N}_{g+1,b}$ is induced by the inclusion $(\mathbf{RP}^2)^{\#g} \rightarrow (\mathbf{RP}^2)^{\#g+1}$ of connected summands.

⁴By this we mean the existence of a dashed lift together with a homotopy γ witnessing the commutativity (up to homotopy) of the diagram.

Warning 3.1.5 (The case $n = 0$). Note by our above definition, it is *not* necessarily the case that $BK(n)$ denotes the classifying space of $K(n)$. In particular, $BSO(0) \simeq S^0$ (encoding the fact that a 0-dimensional real vector space admits two orientations), while the classifying space of $SO(0) = 0$ is contractible.

Given $n \geq 1$ and $K = O, SO$, we fix a well-known model for the classifying space $BK(n)$ and the corresponding tautological n -plane bundle $\gamma_n^{K(n)}$ over $BK(n)$, classifying vector bundles with $K(n)$ -structure.

1. We define $BO(n) := \text{Gr}_n(\mathbf{R}^\infty)$, the infinite Grassmannian of n -planes in $\mathbf{R}^\infty$. The total space U_n^O of γ_n^O is defined so that the fiber over the n -plane $V \in BO(n)$ is itself, topologized as a subspace of the product $BO(n) \times \mathbf{R}^\infty$ and equipped with the obvious projection map.
2. $BSO(n) := \text{Gr}_n^+(\mathbf{R}^\infty)$, the infinite Grassmannian of oriented n -planes in $\mathbf{R}^\infty$. The tautological n -plane bundles γ_n^{SO} is defined analogously to the unoriented case.
3. We define $EK(n) := \text{Fr}^K(U_n^K)$, the frame bundle associated to the tautological bundle γ_n^K .

3.2. Thom spectra.

We denote by $\text{Ho}(\mathbf{Sp})$ the stable homotopy category of Bousfield and Friedlander. It is given by the homotopy category of a symmetric monoidal model category of spectra $(\mathbf{Sp}, \wedge, \mathbf{S}^0)$, of which there are several (Quillen-)equivalent models [MMSS01]. Since we are only interested in constructions within the stable homotopy category, for simplicity we implicitly work in the category of sequential spectra, where the weak equivalences are given by π_* -isomorphisms, cofibrations are given levelwise via the Quillen model structure on Top_* , and fibrations are defined via a suitable right lifting property.

Definition 3.2.1. Given a vector bundle V on a CW-complex X , we define the *Thom spectrum* X^V of V as the suspension spectrum of the Thom space $\text{Th}(V)$ of V :

$$X^V := \Sigma^\infty \text{Th}(V).$$

We extend this Thom spectrum construction to virtual vector bundles as follows. If X is compact (e.g., a finite complex), every virtual vector bundle $\alpha = E_1 - E_2$ on X can be expressed as the difference $E - \mathbf{n}_X$ of a genuine vector bundle E and a trivial vector bundle of sufficiently high rank. That is, we have an isomorphism of vector bundles $E_1 \oplus \mathbf{n}_X \cong E \oplus E_2$. We define the *Thom spectrum* X^α of α by

$$X^\alpha := \Sigma^{-n} X^E.$$

More generally, if X can be filtered by compact subcomplexes X_n , we can extend the above construction by pulling back the vector bundle V to a bundle $V_n := V|_{X_n}$, and defining X^V to be the (homotopy) colimit of the filtered system of Thom spectra $X_n^{V_n}$ ⁵. For example, if X has a CW-structure with finitely many cells in each dimension, the skeleta $X^{(n)}$ of X yield such a filtration, and indeed this will be the case for all of the spaces we consider.

Definition 3.2.2. Given a continuous map $f: Y \rightarrow X$, the *pullback* $f^*(\alpha)$ of α along f is the virtual vector bundle over Y defined by

$$f^*(\alpha) := f^*(E_1) - f^*(E_2).$$

3.3. Madsen-Tillmann spectra and Madsen-Tillmann-Weiss maps.

We are now ready to define the Madsen-Tillman spectra, and relate them to the stable oriented and non-orientable mapping class groups.

Definition 3.3.1. Given $n \geq 0$, the *Madsen-Tillmann spectrum* $MTK(n)$ is defined as the Thom spectrum of the virtual vector bundle $-\gamma_n^K$ over $BK(n)$.

It is not immediately obvious from the definition given how we can identify $\mathbf{Z} \times B\Gamma_\infty^+$ and $B\mathcal{N}_\infty^+$ as infinite loop spaces of Madsen-Tillmann spectra. The following construction will serve as the necessary intermediary step to bridge the two seemingly disparate mathematical objects.

Let $M \rightarrow E \xrightarrow{f} B$ be a smooth fiber bundle with fiber M^n a closed n -manifold with $K(n)$ -structure. In particular, f is a surjective submersion, hence we have a short exact sequence of vector bundles on E

$$(3.2) \quad 0 \rightarrow T_f E \rightarrow TE \rightarrow f^*TB \rightarrow 0$$

⁵Given a map $X_i \rightarrow X_j$ in the filtered system $X = \text{hocolim}_n X_n$, the map $X_i \rightarrow X$ factors through X_j and so we. Functoriality of the Thom space construction yields a map $X_i^{V_i} \rightarrow X_j^{V_j}$ which form the structure maps of our filtered system of Thom spectra.

where the kernel $T_f E$ is called the *vertical or (fiberwise) tangent bundle* of f . Note $T_f E$ naturally inherits a $K(n)$ -structure, hence it is classified by a map $\psi: E \rightarrow BK(n)$, unique up to homotopy. This yields a map at the level of Thom spectra:

$$\kappa_E: E^{-T_f E} \rightarrow MTK(n).$$

Fix a smooth embedding $i: E \hookrightarrow \mathbf{R}^n$. From this, we define a fiberwise inclusion $j = (f, i): E \rightarrow \underline{n}_B$ with normal bundle ν_j . We have the following short exact sequence of vector bundles on E :

$$(3.3) \quad 0 \rightarrow TE \rightarrow f^*TB \oplus \underline{n}_E \rightarrow \nu_j \rightarrow 0.$$

Combining the short exact sequences 3.2 and 3.3 above yields a new short exact sequence

$$0 \rightarrow T_f E \rightarrow \underline{n}_E \rightarrow \nu_j \rightarrow 0,$$

hence we get an identification of virtual vector bundles $-T_f E = \nu_j - \underline{n}_E$. The Pontryagin-Thom collapse applied to j yields a map of Thom spectra

$$\Sigma^n f^!: \Sigma_+^{\infty+n} B \rightarrow \Sigma^n E^{-T_f E}.$$

Definition 3.3.4. We define the *Madsen-Tillmann-Weiss map* (or *MTW map*) α_E of the bundle f as the composite

$$\alpha_E: \Sigma_+^{\infty} B \xrightarrow{f^!} E^{-T_f E} \xrightarrow{\kappa_E} MTK(n).$$

Example 3.3.5. The canonical embedding $i: K(n) \hookrightarrow K(n+1)$ induces, via Proposition 3.1.3, an n -sphere bundle $S^n \rightarrow BK(n) \xrightarrow{Bi} BK(n+1)$. We denote by α the MTW map

$$\alpha: \Sigma_+^{\infty} BK(n+1) \rightarrow MTK(n)$$

associated to Bi .

Given $g > 0$, $B\Gamma_g$ classifies smooth bundles with fiber Σ_g (and orientation-preserving transition maps). Let α_g denote the map

$$\alpha_g: \mathbf{Z} \times B\Gamma_g \rightarrow \Omega^{\infty} MTSO(2)$$

given taking the adjoint of the MTW map applied to the universal Σ_g -bundle over $B\Gamma_g$.⁶ Taking $g \rightarrow \infty$ gives us an analogous map α_{∞}^{SO} . In the landmark paper resolving the Mumford conjecture [Ref], Madsen and Weiss showed that α_{∞}^{SO} induced an integral homology equivalence. Combined with the fact that $\pi_1 MTSO(2) \cong 0$, this implies we can extend α_{∞} to a homotopy equivalence

$$\alpha_{\infty}^{SO}: \mathbf{Z} \times B\Gamma_{\infty}^+ \xrightarrow{\sim} \Omega^{\infty} MTSO(2).$$

The identification of the (plus construction of the) non-orientable stable mapping class group with the infinite loop space of a Madsen-Tillmann spectra is almost completely analogous, and results in a homotopy equivalence between the plus construction of the classifying space of the non-orientable infinite mapping class group and the basepoint component of the infinite loop space of the unoriented Madsen-Tillmann spectrum $MTO(2)$

$$\alpha_{\infty}^O: B\mathcal{N}_{\infty}^+ \xrightarrow{\sim} \Omega_0^{\infty} MTO(2).$$

3.3.1. A cofiber sequence of Madsen-Tillman spectra. There are two more maps involving Madsen-Tillmann spectra we would like to define. With our Grassmannian model for $BK(n)$, $n \geq 1$, the natural inclusion $\mathbf{R}^k \rightarrow \mathbf{R}^{k+1}$ of the first k -coordinates induces an embedding $\mathrm{Gr}_n(\mathbf{R}^k) \hookrightarrow \mathrm{Gr}_n(\mathbf{R}^{k+1})$ (and similarly in the oriented case). Passing to the colimit as $k \rightarrow \infty$ gives an embedding $j: BK(n) \rightarrow BK(n+1)$ as well as a bundle isomorphism $U_{n+1}^K|_{BK(n)} \cong U_n^K \oplus \underline{1}$. That is, we have a pullback square

$$\begin{array}{ccc} U_n^K \oplus \underline{1} & \longrightarrow & U_{n+1}^K \\ \gamma_n^K \oplus \underline{1} \downarrow & \lrcorner & \downarrow \gamma_{n+1}^K \\ BK(n) & \xrightarrow{j} & BK(n+1) \end{array}$$

⁶Strictly speaking, this yields a (based) map $\alpha'_g: B\Gamma_g \rightarrow \Omega^{\infty} MTSO(2)$. Note $\pi_0(MTSO(2)) \cong \mathbf{Z}$, and all the components of an infinite loop space are homotopy equivalent, hence we have a (non-canonical) equivalence $\Omega^{\infty}(MTSO(2)) \simeq \mathbf{Z} \times \Omega_0^{\infty}(MTSO(2))$. This leads to a well-defined map (up to homotopy) α_g^{SO} defined on the component $\{k\} \times B\Gamma_g$ by $\{k\} \times \alpha'_g$, for every $k \in \mathbf{Z}$.

Passing to Thom spectra induces a map

$$\eta: MTK(n) \rightarrow \Sigma MTK(n+1).$$

Note η is the virtual analog of the n^{th} structure map of the Thom spectrum MK .

Given $n \geq 1$, since $BK(n)$ is a CW-complex, U_n^K is a direct summand of a trivial bundle $U_n^K \oplus E \cong \underline{m}_{BK(n)}$, hence $MTK(n) \simeq \Sigma^{-m} BK(n)^E$. The bundle inclusion $E \hookrightarrow \underline{m}_{BK(n)}$ induces a map

$$\beta: MTK(n) \rightarrow \Sigma_+^\infty BK(n),$$

unique up to homotopy.

The following is our main computational tool for computing homotopy groups of Madsen-Tillmann spectra.

Proposition 3.3.6 (Proposition 3.1 of [GMTW09]). *We have a cofiber sequence*

$$MTK(n+1) \xrightarrow{\beta} \Sigma_+^\infty BK(n+1) \xrightarrow{\alpha} MTK(n) \xrightarrow{\eta} \Sigma MTK(n+1) .$$

4. THE STABLE HOMOTOPY CATEGORY

Let $\mathbf{Sp} = (\mathbf{Sp}, \wedge, \mathbf{S}^0)$ denote one of the various (Quillen) equivalent symmetric monoidal model categories of spectra, as in [MMSS01].

Definition 4.0.1. We say a spectrum E is of *finite type* if E has the homotopy type of a finite CW spectrum.

Definition 4.0.2. We say a graded R -module M is of *finite type* if M_n is finitely generated in each degree n .

In any category with arbitrary products, coproducts and a zero object (for example, a pointed model category), given a collection of objects X_α , we have a natural comparison map

$$\varphi: \bigvee_\alpha X_\alpha \rightarrow \prod_\alpha X_\alpha$$

by defining the component map $\varphi_{\beta\alpha}: X_\alpha \rightarrow \bigvee_\alpha X_\alpha \rightarrow \prod_\alpha X_\alpha \rightarrow X_\beta$ to be the identity if $\beta = \alpha$ and the zero map otherwise.

Proposition 4.0.3 (Proposition 3.14 of [Ada74]). *If for each n , $\pi_n(X_\alpha) = 0$ for all but finitely many α , then the above map is a (weak) homotopy equivalence.*

Proof. Since stable homotopy is a homology theory, $\pi_*(-)$ takes arbitrary wedges in the stable homotopy category to direct sums. By definition of stable homotopy groups, as in the unstable case, $\pi_*(-)$ takes arbitrary products to direct products. Thus, for each n ,

$$\pi_n \left(\bigvee_\alpha X_\alpha \right) \cong \bigoplus_\alpha \pi_n(X_\alpha) \cong \prod_\alpha \pi_n(X_\alpha) = \pi_n \left(\prod_\alpha X_\alpha \right)$$

as finite products coincide with finite coproducts in the category of abelian groups, and the isomorphism of abelian groups is by definition induced by $\pi_n(\varphi)$ by functoriality. $\square$

5. LOCALIZATION AND COMPLETION OF SPECTRA

Fix a prime p . Throughout, we will be working within the homotopy category of spectra localized at/localized away from/or completed at p . All three of these alterations to the classical stable homotopy category can be formalized through the theory of (left) Bousfield localization. We briefly recount the theory of localizations and completions of spectra, following [BR20]. Another excellent resource on this subject is [TL22].

Let A be an abelian group. In particular, we are interested in the cases A is one of $\mathbf{Z}_{(p)}$, $\mathbf{Z}_p$, or $\mathbf{Z}/p$. We can resolve A by a short exact sequence of free abelian groups

$$0 \rightarrow F_2 \xrightarrow{\varphi} F_1 \rightarrow A \rightarrow 0$$

where $F_1 = \mathbf{Z}\{I_1\}$ can be thought of as encoding the generators of A , and $F_2 = \mathbf{Z}\{I_2\}$ encoding the relations. From this we can define a map of spectra $\tilde{\varphi}: \bigvee_{I_2} \mathbf{S}^0 \rightarrow \bigvee_{I_1} \mathbf{S}^0$ which induces the homomorphism φ on π_0 . The uniquely characterizes $\tilde{\varphi}$ up to homotopy.

Definition 5.0.1. We define the *Moore spectrum* $M(A)$ of A to be the cofiber of the map $\tilde{\varphi}$.

$$M(A) := \text{hocofib}(\tilde{\varphi}).$$

The Moore spectrum of A is well-defined up to (weak) homotopy type. Indeed, by the stable Hurewicz theorem, along with the long exact sequence on homology associated to a cofiber sequence, $M(A)$ is characterized by its singular (co)homology

$$H_n(M(A), \mathbf{Z}) \cong \begin{cases} A, & n = 0 \\ 0, & o/w. \end{cases}.$$

The p -localized and p -complete stable homotopy categories arise as the homotopy category of the *left Bousfield localizations* of the model category of spectra with respect to the class of $M(A)$ -equivalences, i.e. those maps of spectra which induce isomorphisms on $M(A)$ -homology, for suitable choices of A .

5.1. Localizations of model categories. For a general simplicial model category $\mathbf{M}$ and a class $\mathcal{S}$ of morphisms in $\mathbf{M}$, the *left Bousfield localization of $\mathbf{M}$ at $\mathcal{S}$* , denoted $L_{\mathcal{S}}\mathbf{M}$, consists of a new model structure on the category underlying $\mathbf{M}$, in which

- cofibrations remain the same,
- weak equivalences are given by $\mathcal{S}$ -local equivalences (defined below),
- fibrations are characterized by a suitable right lifting property against trivial cofibrations.

The class $\mathcal{W}_{\mathcal{S}}$ of $\mathcal{S}$ -local equivalences in $\mathbf{M}$ are defined in two steps.

Definition 5.1.1. An object X in $\mathbf{M}$ is $\mathcal{S}$ -local if the presheaf

$$\mathbf{R}\text{map}_{\mathbf{M}}(-, X) : \mathbf{M}^{\text{op}} \rightarrow \text{Ho}(\mathbf{sSet})$$

sends morphisms in $\mathcal{S}$ to weak equivalences (with respect to the Quillen model structure on the category of simplicial sets).

Intuitively, an $\mathcal{S}$ -local object is an object in $\mathbf{M}$ which “sees” the morphisms in $\mathcal{S}$ as isomorphisms.

Definition 5.1.2. A morphism $f : A \rightarrow B$ in $\mathbf{M}$ is an $\mathcal{S}$ -local equivalence if for every $\mathcal{S}$ -local object X in $\mathbf{M}$, the map

$$f^* : \mathbf{R}\text{map}_{\mathbf{M}}(B, X) \rightarrow \mathbf{R}\text{map}_{\mathbf{M}}(A, X)$$

is a weak equivalence of Kan complexes.

Clearly based on the definition of $\mathcal{S}$ -local objects we have the inclusion $\mathcal{S} \subset \mathcal{W}_{\mathcal{S}}$. Somewhat more surprisingly, Theorem 17.6.3 in [Hirschhorn] implies that $\mathcal{W}_{\mathcal{S}}$ also contains the weak equivalences of $\mathbf{M}$.

It follows that the identity functors

$$\text{Id}_{\mathbf{M}} : \mathbf{M} \rightleftarrows L_{\mathcal{S}}\mathbf{M} : \text{Id}_{L_{\mathcal{S}}\mathbf{M}}$$

form a Quillen pair. The associated left derived functor $L_{\mathcal{S}} := \mathbf{L}\text{Id}_{\mathbf{M}}$ is called the *Bousfield localization functor*. Together the pair $(L_{\mathcal{S}}\mathbf{M}, \text{Id}_{\mathbf{M}})$ satisfy the following universal property: any other model category $\mathbf{N}$ and left Quillen functor $F : \mathbf{M} \rightarrow \mathbf{N}$ which sends the morphisms of $\mathcal{S}$ to weak equivalences factors uniquely through a left Quillen functor out of $L_{\mathcal{S}}\mathbf{M}$:

$$\begin{array}{ccc} \mathbf{M} & \xrightarrow{F} & \mathbf{N} \\ \text{Id}_{\mathbf{M}} \downarrow & \nearrow \exists! & \\ L_{\mathcal{S}}\mathbf{M} & & \end{array}.$$

Thus in particular, left Bousfield localizations are unique up to isomorphism if they exist. In addition, the associated right derived functor $i := \mathbf{R}\text{Id}_{L_{\mathcal{S}}\mathbf{M}} : \text{Ho}(L_{\mathcal{S}}\mathbf{M}) \rightarrow \text{Ho}(\mathbf{M})$ is fully faithful, and hence identifies $\text{Ho}(L_{\mathcal{S}}(\mathbf{M}))$ with the full subcategory of $\text{Ho}(\mathbf{M})$ given by the essential image of i . One is then led to ask if the objects which span this subcategory have a nice characterization, and indeed they do.

Proposition 5.1.3. *The essential image of the right adjoint $i : \text{Ho}(L_{\mathcal{S}}\mathbf{M}) \rightarrow \text{Ho}(\mathbf{M})$ is the subcategory spanned by $\tilde{\mathcal{S}}$ -local objects, where $\tilde{\mathcal{S}}$ is the image of $\mathcal{S}$ under the homotopy functor $\gamma : \mathbf{M} \rightarrow \text{Ho}(\mathbf{M})$.*

Proof. By construction, the Bousfield localization of $\mathbf{M}$ at $\mathcal{S}$ precisely induces a localization of the homotopy category $\mathrm{Ho}(\mathbf{M})$ at the class of morphisms $\tilde{\mathcal{S}} = \gamma(\mathcal{S})$. By the above universal property, the localization functor $L_{\mathcal{S}}: \mathrm{Ho}(\mathbf{M}) \rightarrow \mathrm{Ho}(L_{\mathcal{S}}\mathbf{M})$ admits a fully faithful right adjoint, hence it is a reflective localization in the ordinary categorical sense. It immediately follows that the right adjoint is a reflection onto the full subcategory of $\gamma(\mathcal{S})$ -local objects, as required. $\square$

Unfortunately, theorems proving the existence of Bousfield localizations often require extra assumptions on the model category $\mathbf{M}$ in question (for example, being *combinatorial* in the sense of Definition 10.8 of [TL22]), accounting for the fact that $\mathcal{S}$ (and hence $\mathcal{W}_{\mathcal{S}}$) are rarely proper sets.

If however $\mathbf{M}$ is a stable model category (such as in the case $\mathbf{M} = \mathbf{Sp}$), its homotopy category is triangulated, and so under some very mild assumptions, we can develop the theory of Bousfield localizations of $\mathbf{M}$ directly at the level of the homotopy category $\mathrm{Ho}(\mathbf{M})$. The major upshot the triangulated structure one inherits from the stable model structure on $\mathbf{M}$ is that given a class of morphisms $\mathcal{S}$ in $\mathrm{Ho}(\mathbf{M})$ with which we want to localize with respect to, we can. In this case, the construction reduces to a special case of the theory of *Verdier localizations*.

5.2. Localizations of triangulated categories.

Definition 5.2.1. Let $\mathcal{T}$ be a triangulated category. A *triangulated subcategory* of $\mathcal{T}$ consists of a nonempty full subcategory $\mathcal{S}$ of $\mathcal{T}$ which satisfies the following properties:

(TS1). The shift functor $\Sigma: \mathcal{T} \xrightarrow{\cong} \mathcal{T}$ restricts to an auto-equivalence on $\mathcal{S}$.

(TS2). Given a cofiber sequence $X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$ in $\mathcal{T}$, if X and Y are contained in $\mathcal{S}$, then so Z .

A triangulated subcategory $\mathcal{S}$ is *thick* if in addition, it is closed under retracts contained in $\mathcal{T}$.

A triangulated subcategory $\mathcal{S}$ of $\mathcal{T}$ inherits a canonical triangulated structure making the inclusion functor $\mathcal{S} \hookrightarrow \mathcal{T}$ a triangulated functor (i.e., an additive functor that commutes with shift functors and preserves cofiber sequences).

Definition 5.2.2. Given an additive functor $F: \mathcal{T} \rightarrow \mathbf{A}$ (whose target is not necessarily triangulated), the *kernel* $\ker F$ of F is the full subcategory of $\mathcal{T}$ spanned by objects X in $\mathcal{T}$ such that $F(X) \cong 0$.

Note the kernel of an additive functor on a triangulated category is always a triangulated subcategory. The kernel of a triangulated functor between triangulated categories is, in addition, a thick subcategory of the source.

Definition 5.2.3. Let $\mathcal{S}$ be a triangulated subcategory of $\mathcal{T}$. We denote by $\Sigma(\mathcal{S})$ the collection of morphisms in $\mathcal{T}$ whose cofiber lies in $\mathcal{S}$.

Definition 5.2.4. The *Verdier localization* $\mathcal{T}/\mathcal{S}$ of a triangulated category $\mathcal{T}$ with respect to a triangulated subcategory $\mathcal{S}$, is the localization of $\mathcal{T}$ at the class of morphisms whose cofiber lies in $\mathcal{S}$:

$$\mathcal{T}/\mathcal{S} := \mathcal{T}[\Sigma(\mathcal{S})^{-1}].$$

A sufficient condition for the existence of a Verdier localization is that $\mathcal{S}$ is a thick subcategory closed under coproducts in $\mathcal{T}$ and $\mathcal{T}$ (and hence $\mathcal{S}$) has all small coproducts. The triangulated structure also endows the localization $\mathcal{T} \rightarrow \mathcal{T}/\mathcal{S}$ with several nice properties.

Proposition 5.2.5. *Let $\mathcal{S} \subset \mathcal{T}$ as above.*

1. *The category $\mathcal{T}/\mathcal{S}$ inherits a unique triangulated structure making the localization functor $Q: \mathcal{T} \rightarrow \mathcal{T}/\mathcal{S}$ exact.*
2. *A morphism in $\mathcal{T}$ is sent to the zero morphism by Q iff the morphism factors through $\mathcal{S}$.*
3. *The kernel of Q is the smallest thick subcategory containing $\mathcal{S}$.*

It is a nontrivial result (Proposition 8.8 of [TL22]) that given a class of morphisms $\mathcal{S} \subset \mathbf{Sp}$, under suitable conditions (including the following cases), the Bousfield localization of $\mathbf{Sp}$ with respect to $\mathcal{S}$ can be realized as a Verdier localization of $\mathrm{Ho}(\mathbf{Sp})$ at a thick subcategory.

Definition 5.2.6. Let X be a spectrum.

1. We refer to $\mathbf{Sp}_p^\wedge := L_{M(\mathbf{Z}/p)}\mathbf{Sp}$ as the *p-complete category of spectra*. The $M(\mathbf{Z}/p)$ -local spectrum $L_{M(\mathbf{Z}/p)}(X)$ is called the *p-completion* of X , and is denoted by $X_p^\wedge$.
2. We refer to $\mathbf{Sp}_{(p)} := L_{M(\mathbf{Z}_{(p)})}\mathbf{Sp}$ as the *p-localized category of spectra*. The $M(\mathbf{Z}_{(p)})$ -local spectrum $L_{M(\mathbf{Z}_{(p)})}(X)$ is called the *localization* of X at p , and is denoted by $X_{(p)}$.

3. Finally, we refer to $\mathrm{Sp}[\frac{1}{p}] := L_{M(\mathbf{Z}_p)}\mathrm{Sp}$ as the p -inverted category of spectra. The $M(\mathbf{Z}_p)$ -local spectrum $L_{M(\mathbf{Z}_p)}(X)$ is called the *localization of X away from p* , and is denoted by $X[\frac{1}{p}]$.

Note the latter two localizations are *smashing*, i.e. in both cases the localization functor is given by smashing with the localization of the sphere spectrum:

$$L_{M(A)}(X) = X \wedge L_{M(A)}(\mathbf{S}^0).$$

While p -completions are not smashing localizations, under some mild finiteness assumptions they do have a similarly nice algebraic characterization.

Proposition 5.2.7. *Fix a spectrum X .*

1. *The p -completion of X is given by the homotopy limit*

$$X_p^\wedge \simeq \mathrm{holim}_n (M(\mathbf{Z}/p^n) \wedge X)$$

where the connecting maps $M(\mathbf{Z}/p^{n+1}) \rightarrow M(\mathbf{Z}/p^n)$ are induced by the projection homomorphisms $\mathbf{Z}/p^{n+1} \rightarrow \mathbf{Z}/p^n$.

2. *If $\pi_*(X)$ is of finite type, the homotopy groups of $X_p^\wedge$ are given by*

$$\pi_*(X_p^\wedge) \cong \pi_*(X) \otimes_{\mathbf{Z}} \mathbf{Z}_p^\wedge.$$

We now fix some terminology related to splitting spectra. The language we use reflects the situation in a general additive category (e.g., a triangulated category), such as the familiar case of modules over a commutative ring.

Definition 5.2.8. A *splitting* of a spectrum E is a homotopy equivalence $f: E_1 \vee E_2 \xrightarrow{\sim} E$. Given such a splitting f , we say E *splits* and E_1 and E_2 *split off* E .

In practice, one often begins with a map of spectra $i: E_1 \rightarrow E$, and asks whether E_1 splits off E . As in the case of modules, a splitting of a spectrum coincides with a split short exact sequence.

Lemma 5.2.9.

1. *Given a map of spectra $s: E_1 \rightarrow E$, E_1 splits off E if and only if there exists a map $r: E \rightarrow E_1$ such that rs is a self-homotopy equivalence.*
2. *Given a map of spectra $r: E \rightarrow E_2$, E_2 splits off E if and only if there exists a map $s: E_2 \rightarrow E$ such that rs is a self-homotopy equivalence.*

Proof. In the stable homotopy category this amounts to a routine verification that in an additive category, a splitting of an object coincides with a split short exact sequence of its summands. In the first case, we have a homotopy equivalence $E \simeq E_1 \vee \mathrm{cofib}(s)$. In the second case, we have $E \simeq E_2 \vee \mathrm{fib}(r)$. $\square$

6. THE (MOD p) STEENROD ALGEBRA

The study of the cohomology of spaces is greatly aided by the existence of cohomology operations. Spectra provide a nice categorical setting to formulate the theory of cohomology operations. As the stable homotopy category is a triangulated category, given a spectrum E , the graded abelian group $E^*(E) = [E, E]_{-*}$ is in fact a graded ring under composition. In addition, the canonical auto-equivalences on the stable homotopy category and the Yoneda embeddings yield the following commutative square

$$\begin{array}{ccc} [E, E]_{-n} = [E, \Sigma^n E] & \longrightarrow & \mathrm{Nat}([- , E], [- , \Sigma^n E]) \\ \Sigma^k \downarrow \cong & & \downarrow (\Sigma^{-k})^* \cdot (-) \cdot \Sigma_*^k \\ [\Sigma^k E, \Sigma^{n+k} E] & \longrightarrow & \mathrm{Nat}([- , E^k], [- , \Sigma^{n+k} E]) \end{array}$$

Thus $[E, E]_{-n}$ parametrizes stable E -cohomology operations of the form $\theta: E^*(-) \rightarrow E^{*+n}(-)$.

Thus, given another spectrum X , the cohomology group $E^*(X)$ is a graded $E^*(E)$ -module as follows. Given (homotopy classes of) maps of spectra $\theta: E \rightarrow \Sigma^n E$ and $f: X \rightarrow \Sigma^m E$, the composite

$$X \xrightarrow{f} \Sigma^m E \xrightarrow{\Sigma^m \theta} \Sigma^{n+m} E$$

represents a well-defined homotopy class in $E^{n+m}(X) = [X, E]_{-n-m} \cong [X, \Sigma^{n+m} E]$. This assignment defines the aforementioned $E^*(E)$ -module structure on $E^*(X)$.

Fix a prime p . Let $H\mathbf{Z}/p$ denote the mod p Eilenberg MacLane spectrum representing singular cohomology with coefficients $\mathbf{Z}/p$. Then the ring of stable cohomology operations $\mathcal{A} = \mathcal{A}_p^* := H\mathbf{Z}/p^*(H\mathbf{Z}/p)$ is known as the *mod p Steenrod algebra*. Since $\mathbf{Z}/p$ is a commutative ring, $H\mathbf{Z}/p$ is a homotopy commutative ring spectrum, hence it comes equipped with multiplication and identity morphisms $\mu: H\mathbf{Z}/p \wedge H\mathbf{Z}/p \rightarrow H\mathbf{Z}/p$ and $\eta: \mathbf{S}^0 \rightarrow H\mathbf{Z}/p$ which fit into the necessary commutativity diagrams to make $H\mathbf{Z}/p$ a commutative monoid object in the stable homotopy category (in fact, given a commutative ring R , HR satisfies further compatibility conditions, making it an E_∞ -ring spectrum, i.e., a commutative monoid in the category of spectra itself).

It is an algebra over $\mathbf{Z}/p$, with the ring inclusion $\mathbf{Z}/p \rightarrow \mathcal{A}$ induced by applying $\pi_*(-)$ to the composite map of spectra

$$H\mathbf{Z}/p \rightarrow F(H\mathbf{Z}/p, H\mathbf{Z}/p \wedge H\mathbf{Z}/p) \xrightarrow{\mu_*} F(H\mathbf{Z}/p, H\mathbf{Z}/p),$$

where the first map is the unit of the smash product internal function spectrum adjunction in the stable homotopy category

For the rest of this section we fix $p = 2$.

The following can be taken as an explicit axiomatic characterization of the mod 2 Steenrod algebra.

Theorem 6.0.1 (Serre-Cartan). *The Steenrod algebra $\mathcal{A}$ is generated (as an algebra) by cohomology operations*

$$Sq^n: H\mathbf{Z}/p \rightarrow \Sigma^n H\mathbf{Z}/p$$

for $n \geq 0$, called the Steenrod squares. Given an arbitrary space X , they are characterized by the following properties:

1. Sq^0 is the identity.
2. Sq^1 is the Bockstein homomorphism β corresponding to the short exact sequence

$$0 \rightarrow \mathbf{Z}/2 \rightarrow \mathbf{Z}/4 \rightarrow \mathbf{Z}/2 \rightarrow 0.$$

3. Given $x \in \tilde{H}^n(X; \mathbf{Z}/2)$, $Sq^n(x) = x^2$.
4. Given $x \in \tilde{H}^m(X; \mathbf{Z}/2)$ and $n > m$, $Sq^n(x) = 0$.
5. (Cartan formula).

$$Sq^n(xy) = \sum_{i+j=n} Sq^i(x)Sq^j(y).$$

6. (Adem relations). Given $0 < n < 2m$,

$$Sq^n Sq^m = \sum_{i=0}^{\lfloor n \rfloor} \binom{m-1-i}{n-2i} Sq^{n+m-i} Sq^i.$$

The fact that $\mathcal{A}$ is generated as an algebra over $\mathbf{Z}/2$ by the Steenrod squares implies that it is generated as a $\mathbf{Z}/2$ -vector space by finite products of the form $Sq^{i_1} \dots Sq^{i_n}$, $n \geq 0$. The existence of Adem relations (property 6) implies we need not consider all such finite products to obtain a generating set

Definition 6.0.2. A monomial $Sq^{i_1} \dots Sq^{i_n}$ in $\mathcal{A}$ is *admissible* if $i_{k-1} \geq 2i_k$ for all $2 \leq k \leq n$.

Proposition 6.0.3. *The admissible monomials form a vector space basis for $\mathcal{A}$.*

Definition 6.0.4. Let $I(\mathcal{A})$ denote the ideal of positive degree elements of the Steenrod algebra (the notation will be justified shortly). An element $Q \in I(\mathcal{A})$ is said to be *decomposable* if it lies in the image of the restricted multiplication map $m: I(\mathcal{A}) \otimes_{\mathbf{Z}/2} I(\mathcal{A}) \rightarrow I(\mathcal{A})$ for $\mathcal{A}$, otherwise we say Q is *indecomposable*.

It is not difficult to verify that given a generating set I for an algebra $\mathcal{R}$, the linear span of I contains all of the indecomposable elements of $\mathcal{R}$.

Proposition 6.0.5. *The indecomposable elements of $\mathcal{A}$ are the Steenrod squares Sq^{2^k} for $k \geq 0$. Thus the Steenrod algebra is generated by the set $\{Sq^{2^k} : k \geq 0\}$.*

Definition 6.0.6. A Hopf algebra over a ring R consists of an R -algebra A together with

1. An R -algebra homomorphism $\varepsilon: A \rightarrow R$ called an *augmentation*, the kernel of which is a two-sided ideal in A called the *augmentation ideal*, and denoted $I(A)$.
2. An R -algebra homomorphism $\Delta: A \rightarrow A \otimes_R A$.
3. An R -linear map $c: A \rightarrow A$ called an antipode.

This data is subject to the following compatibility requirements:

Theorem 6.0.7. *The Steenrod algebra $\mathcal{A}$ is a non-commutative Hopf algebra over $\mathbf{Z}/2$ with cocommutative comultiplication defined by*

$$\Delta(Sq^n) := \sum_{i=0}^n Sq^i \otimes Sq^{n-i}$$

and antipode defined inductively by

$$c(Sq^0) = Sq^0, \quad \sum_{i=0}^n Sq^i c(Sq^{n-i}) = 0 \quad \text{for } n \geq 1.$$

One can also look at the dual Steenrod algebra

$$\mathcal{A}_* := H\mathbf{Z}/2_*(H\mathbf{Z}/2) \cong \pi_*(H\mathbf{Z}/2 \wedge H\mathbf{Z}/2),$$

also known as the *coalgebra of cooperations* in mod p homology. By the universal coefficient theorem [Adams74 proposition 13.5], we have an isomorphism of graded $\mathbf{Z}/2$ -vector spaces

$$\mathcal{A} = H\mathbf{Z}/2^*(H\mathbf{Z}/2) \cong \text{Hom}_{\mathbf{Z}/2}^*(H\mathbf{Z}/2_*(H\mathbf{Z}/2), \mathbf{Z}/2) = \mathcal{A}_*^\vee.$$

Since $\mathcal{A}_*$ is of finite type, we get canonical isomorphisms

$$\mathcal{A}_* \cong (\mathcal{A}_*^\vee)^\vee \cong \mathcal{A}^\vee$$

i.e., $\mathcal{A}$ and $\mathcal{A}_*$ are dual as graded $\mathbf{Z}/2$ -vector spaces. Thus the Hopf algebra structure on $\mathcal{A}$ dualizes to make $\mathcal{A}_*$ a commutative, but not cocommutative, Hopf algebra over $\mathbf{Z}/2$.

One upside of considering the dual Steenrod algebra $\mathcal{A}_*$ is that it is a commutative algebra, hence its Hopf algebra structure can be more easily described in terms of the commutative ring spectrum structure of $H\mathbf{Z}/2$. Indeed, the multiplication on $\mathcal{A}_*$ is given by the composite

$$\pi_*(H\mathbf{Z}/2 \wedge H\mathbf{Z}/2) \otimes_{\mathbf{Z}/2} \pi_*(H\mathbf{Z}/2 \wedge H\mathbf{Z}/2) \xrightarrow{\Delta} \pi_*(H\mathbf{Z}/2 \wedge H\mathbf{Z}/2) \xrightarrow{\mu_*} \pi_*(H\mathbf{Z}/2 \wedge H\mathbf{Z}/2).$$

In fact, Milnor proved that as far as infinitely generated commutative algebras go, $\mathcal{A}_*$ is as simple as possible.

Theorem 6.0.8 (Serre-Milnor). $\mathcal{A}_* = \mathbf{Z}/2[\xi_1, \xi_2, \dots]$, where $\xi_n: \mathcal{A} \rightarrow \mathbf{Z}/2$ is the dual element of the admissible monomial $Sq^{2^{n-1}} \dots Sq^2 Sq^1$.

7. HOMOLOGICAL ALGEBRA OVER GRADED RINGS

Throughout, we let $\mathcal{R}$ be a (not necessarily commutative) connected (cohomologically) graded algebra over a field k . Furthermore, we assume $\mathcal{R}$ is equipped with a choice of augmentation ε as in definition 6.0.6. The multiplication on k allows us to view k as an $\mathcal{R}$ -algebra and ε as an $\mathcal{R}$ -algebra homomorphism. The spectral sequence arguments we will give require some prerequisite knowledge of the theory of homological algebra within the category $\text{Mod}_{\mathcal{R}}$ of (graded) left $\mathcal{R}$ -modules. The particular case to keep in mind is $\mathcal{R} = \mathcal{A}$, the mod 2 Steenrod algebra.

We say an $\mathcal{R}$ -module homomorphism $f: M \rightarrow N$ is of *degree n* if $f(M^k) \subset N^{k+n}$ for each $n \in \mathbf{Z}$. The category of $\mathcal{R}$ -modules is defined so that the hom-set $\text{Hom}_{\mathcal{R}}(M, N)$ consists of degree 0 maps from M to N .

For each $n \in \mathbf{Z}$, we have a shift auto-equivalence Σ^n on $\text{Mod}_{\mathcal{R}}$ defined by

$$(\Sigma^n M)^k := M^{k-n}, \quad (\Sigma^n f)^k := f^{k-n}.$$

One can readily see that $\text{Hom}_{\mathcal{R}}(\Sigma^n M, N) \cong \text{Hom}_{\mathcal{R}}(M, \Sigma^{-n} N)$ consists of the set of degree n homomorphisms from M to N .

7.1. Ext groups and minimal resolutions. The category of $\mathcal{R}$ -modules is symmetric monoidal with respect to the tensor product $\otimes_k$, and furthermore it is *closed*, in that for each pair of $\mathcal{R}$ -modules M and N , we have an internal-hom $\mathcal{R}$ -module $\underline{\text{Hom}}_{\mathcal{R}}^*(M, N)$, which in degree n consists of degree $-n$ homomorphisms:

$$\underline{\text{Hom}}_{\mathcal{R}}^n(M, N) := \text{Hom}_{\mathcal{R}}(M, \Sigma^n N).$$

This grading convention ensures that (relative to homological grading), a degree n map of spectra $f: \Sigma^n X \rightarrow Y$ yields a degree n element f^* in $\text{Hom}_{\mathcal{A}}^*(H^*(Y), H^*(X))$.

Recall, given $\mathcal{R}$ -modules M and N , we define $\text{Ext}_{\mathcal{R}}^{\bullet}(M, -)$ and $\text{Ext}_{\mathcal{R}}^{\bullet}(-, N)$ to be the right derived functors of the internal hom functors

$$\begin{aligned}\text{Ext}_{\mathcal{R}}^s(M, -) &:= \mathbf{R}^s \underline{\text{Hom}}(M, -): \text{Mod}_{\mathcal{R}} \longrightarrow \text{Mod}_{\mathcal{R}} \\ \text{Ext}_{\mathcal{R}}^s(-, N) &:= \mathbf{R}^s \underline{\text{Hom}}(-, N): \text{Mod}_{\mathcal{R}}^{\text{op}} \longrightarrow \text{Mod}_{\mathcal{R}}.\end{aligned}$$

Thus, for each $s \geq 0$, $\text{Ext}^s(M, N)^*$ is a graded $\mathcal{R}$ -module whose degree t component we write as $\text{Ext}^{s,t}(M, N)$. We refer to s as the *homological degree* and t as the *internal degree*. We can compute

The fact that $\mathcal{R}$ is an augmented algebra implies the existence of a class of projective resolutions in $\text{Ch}(\mathcal{R})$ which make it comparatively easy to compute certain Ext groups of $\mathcal{R}$ -modules.

Definition 7.1.1. We say a projective resolution $(P_{\bullet}, \partial_{\bullet}) \rightarrow M$ of an $\mathcal{R}$ -module M is *minimal* if for every $s \geq 0$,

$$\text{im}(\partial_{s+1}) \subset I(\mathcal{R}) \cdot P_s.$$

Given a minimal projective resolution $P_{\bullet}$ of M , applying $\underline{\text{Hom}}_{\mathcal{R}}(-, k)$ yields a cochain complex

$$\underline{\text{Hom}}_{\mathcal{R}}(P_0, k) \xrightarrow{\partial_1^*} \underline{\text{Hom}}_{\mathcal{R}}(P_1, k) \xrightarrow{\partial_2^*} \dots$$

such that $\partial_s^* = 0$ for all $s \geq 0$. Indeed, this follows immediately from the fact that $I(\mathcal{R}) \cdot k = 0$ by the definition of the $\mathcal{R}$ -module structure of k . In particular, this implies we have an isomorphism of $\mathcal{R}$ -modules

$$\text{Ext}_{\mathcal{R}}^s(M, k) \cong \underline{\text{Hom}}_{\mathcal{R}}(P_s, k).$$

As $\mathcal{R}$ is connected, every $\mathcal{R}$ -module admits a minimal resolution by free $\mathcal{R}$ -modules, i.e., modules of the form $P_s = \bigoplus_{i_s, j_s} \Sigma^{i_s} \mathcal{R}^{\oplus j_s}$, $i_s \geq 0$, $j_s \in \mathbf{N} \cup \{\infty\}$. Furthermore, if M is of finite type, we can ensure each P_s is of finite type as well (i.e. each j_s is finite).

Example 7.1.2. Let $G_3 \subset \mathcal{A}$ denote the vector subspace spanned by admissible monomials Sq^I with $I = (i_1, \dots, i_n)$ an admissible sequence containing at least three positive indices. The Adem relations imply G_3 is an $\mathcal{A}$ -module, and in fact it is an ideal of $\mathcal{A}$. The quotient module $D_2 := \mathcal{A}/G_3$ embeds canonically as a $\mathbf{Z}/2$ -vector subspace of $\mathcal{A}$, with a basis given by all admissible monomials of the form $Sq^{i_2} Sq^{i_1}$. We construct a minimal free resolution $F_{\bullet} \rightarrow D_2$ by $\mathcal{A}$ -modules as follows.

At the first stage we need an epimorphism $F_0 \rightarrow D_2$, which amounts to identifying a generating set of D_2 . By construction, the cokernel $\varepsilon: \mathcal{A} \rightarrow \mathcal{A}/G_2 = D_2$ gives us such an epimorphism, hence we set $F_0 := \mathcal{A}$. We now determine an epimorphism $d_1: F_1 \rightarrow \ker(\varepsilon)$. This amounts to determining a generating set $\{g_{1,i}\}$ for the $\mathcal{A}$ -module $\ker(\varepsilon) = G_2$. We then define F_1 to be the free $\mathcal{A}$ -module $F_1 := \bigoplus_i \Sigma^{|g_{1,i}|} \mathcal{A}\{g_{1,i}\}$, with the epimorphism $d_1: F_1 \rightarrow \ker(d_1)$ defined on the i^{th} -summand by $Sq^0 \mapsto g_{1,i}$. Up to degree 18, G_3 has a generating set given by the second row of the chart 7.1.2 below.

	$\mathcal{A}$ -module type	Image of basis along $d_i: F_i \rightarrow \ker(d_{i-1})$
F_0	$\mathcal{A}$	Sq^0
F_1	$\Sigma^7 \mathcal{A} \oplus \Sigma^{11} \mathcal{A} \oplus \Sigma^{13} \mathcal{A} \oplus \Sigma^{14} \mathcal{A}$	$Sq^4 Sq^2 Sq^1, Sq^8 Sq^2 Sq^1$ $Sq^8 Sq^4 Sq^1 + Sq^9 Sq^3 Sq^1 + Sq^{10} Sq^2 Sq^1,$ $Sq^8 Sq^4 Sq^2 + Sq^9 Sq^4 Sq^1 + Sq^{10} Sq^3 Sq^1 + Sq^{11} Sq^2 Sq^1$
F_2	$\Sigma^{11} \mathcal{A} \oplus \Sigma^{13} \mathcal{A} \oplus \Sigma^{14} \mathcal{A}$	$(Sq^3 Sq^1 + Sq^4)(g_{1,0}), (Sq^4 Sq^2 + Sq^5 Sq^1 + Sq^6)(g_{1,0}) + Sq^2(g_{1,1}),$ $(Sq^4 Sq^2 Sq^1 + Sq^6 Sq^1)(g_{1,0}) + (Sq^2 Sq^1 + Sq^3)(g_{1,1}) + Sq^1(g_{1,2})$
F_3	$\Sigma^{15} \mathcal{A}$	$(Sq^3 Sq^1 + Sq^4)(g_{2,0}) + Sq^2(g_{2,1}) + Sq^1(g_{2,2})$

7.2. The Yoneda product. It is a standard exercise in homological algebra [ref: Weibel] to show that the $\mathcal{R}$ -module $\text{Ext}_{\mathcal{R}}^1(M, N)$ is naturally isomorphic (as a graded abelian group) to $\text{Ext}_{\mathcal{R}}(M, N)$, which in degree t is defined to be the set of isomorphism classes of extensions

$$0 \rightarrow \Sigma^t N \rightarrow E \rightarrow M \rightarrow 0$$

with group structure induced by the Baer sum.

Example 7.2.1. There are several classes in $\text{Ext}_{\mathcal{A}}^{1,*}(\mathbf{Z}/2, \mathbf{Z}/2)$ whose corresponding $\mathcal{A}$ -module extensions can be realized by induced maps on cohomology groups arising from maps of spaces. Thus for example, defining

$M_1 = \mathbf{Z}/2\{x, y\}$ with $|x| = 0$ and $|y| = 1$ connected and a non-trivial square one action $Sq^1(x) = y$, the extension

$$0 \rightarrow \Sigma \mathbf{Z}/2 \xrightarrow{1 \mapsto y} M_1 \rightarrow \mathbf{Z}/2 \rightarrow 0$$

is realized (up to a shift) by the cofiber sequence corresponding to the inclusion of the 1-skeleton in $\mathbf{R}P^2 = S^1 \cup_2 e^2$:

$$0 \rightarrow S^1 \rightarrow \mathbf{R}P^2 \rightarrow S^2 \rightarrow 0 .$$

We represent this situation by the following diagram in which the dots represent copies of $\mathbf{Z}/2$, the horizontal arrows represent $\mathcal{A}$ -module homomorphisms, and the curved vertical arrow represents the nontrivial Steenrod square $Sq^1(w_1) = w_1^2$ in $\tilde{H}^*(\mathbf{R}P^2)$:

$$\begin{array}{ccc} \bullet & \xrightarrow{\quad} & \bullet \\ & \searrow Sq^1 & \\ & \bullet & \xrightarrow{\quad} \bullet \end{array} .$$

$$0 \rightarrow \tilde{H}^*(S^2) \cong \Sigma^2 \mathbf{Z}/2 \rightarrow \tilde{H}^*(\mathbf{R}P^2) \cong \Sigma M_1 \rightarrow \tilde{H}^*(S^1) \cong \Sigma \mathbf{Z}/2 \rightarrow 0$$

The corresponding element in $\text{Ext}_{\mathcal{A}}^{1,1}(\mathbf{Z}/2, \mathbf{Z}/2)$ is denoted by h_0 .

The complex projective space $\mathbf{C}P^2$ and quaternionic projective space $\mathbf{H}P^2$ similarly realize $\mathcal{A}$ -modules admitting nontrivial Sq^2 and Sq^4 actions respectively: letting $M_2 = \Sigma^{-2} \tilde{H}^*(\mathbf{C}P^2)$ and $M_3 = \Sigma^{-4} \tilde{H}^*(\mathbf{H}P^2)$ and denoting in either case the generator in degree 0 by x , we have the following short exact sequences

$$0 \rightarrow \Sigma^2 \mathbf{Z}/2 \xrightarrow{1 \mapsto Sq^2(x)} M_2 \rightarrow \mathbf{Z}/2 \rightarrow 0$$

$$0 \rightarrow \Sigma^4 \mathbf{Z}/2 \xrightarrow{1 \mapsto Sq^4(x)} M_3 \rightarrow \mathbf{Z}/2 \rightarrow 0$$

which correspond to elements $h_1 \in \text{Ext}_{\mathcal{A}}^{1,2}(\mathbf{Z}/2, \mathbf{Z}/2)$ and $h_2 \in \text{Ext}_{\mathcal{A}}^{1,4}(\mathbf{Z}/2, \mathbf{Z}/2)$ respectively.

The Yoneda product is a generalization of this construction to higher Ext groups. Firstly, given $\mathcal{R}$ -modules M and N , a *degree t extension of length s* of M by N consists of an exact sequence of the form

$$0 \rightarrow N \rightarrow E_s \rightarrow E_{s-1} \rightarrow \dots \rightarrow E_1 \rightarrow \Sigma^t M \rightarrow 0 .$$

Given another extension

$$0 \rightarrow L \rightarrow F_{s'} \rightarrow F_{s'-1} \rightarrow \dots \rightarrow F_1 \rightarrow \Sigma^{t'} N \rightarrow 0,$$

we define the corresponding *Yoneda product* as the degree $t' + t$ extension of M by L of length $s' + s$ given by splicing together the two extensions:

$$\begin{array}{ccccccc} 0 \rightarrow L \rightarrow F_{s'} \rightarrow F_{s'-1} \rightarrow \dots \rightarrow F_1 & \xrightarrow{\quad} & \Sigma^{t'} E_s & \rightarrow \dots \rightarrow \Sigma^{t'+t} M \rightarrow 0 \\ & \searrow & \nearrow & \\ & & \Sigma^{t'} N & \end{array} .$$

We can identify $\text{Ext}_{\mathcal{R}}^{s,t}(M, N)$ with suitable equivalence classes of degree t extensions of length s , in which case the above construction induces a pairing of Ext groups

$$\text{Ext}_{\mathcal{R}}^{s,t}(M, N) \otimes_k \text{Ext}_{\mathcal{R}}^{s',t'}(N, L) \rightarrow \text{Ext}_{\mathcal{R}}^{s+s',t+t'}(M, L)$$

known as the *Yoneda product*. Note the Yoneda product admits a direct construction in terms of the right derived functor of the natural composition product

$$\underline{\text{Hom}}_{\mathcal{R}}(M, N) \otimes_k \underline{\text{Hom}}_{\mathcal{R}}(N, P) \rightarrow \underline{\text{Hom}}_{\mathcal{R}}(M, P).$$

8. THE ADAMS SPECTRAL SEQUENCE

The Adams spectral sequence is among the most powerful computational tools in stable homotopy theory. Informally, it is a way of extracting homotopical information (namely, about the graded abelian group of homotopy classes of maps between two spectra) via (co)homological inputs. Slightly more formally, to any homotopy commutative ring spectrum E satisfying a certain *flatness condition* along with a pair of spectra X and Y such that $E_*(X)$ and $E_*(Y)$ are of finite type, one can associate an *E -based Adams spectral sequence*

$$E_2^{s,t} = \text{Ext}_{E_*(E)}(E_*(X), E_*(Y)) \Rightarrow [X, Y_E^\wedge]_{t-s},$$

where $E_*(E) = [E, E]_{-*}$ denotes the coalgebra of stable E -homology cooperations, and $Y_E^\wedge$ denotes the E -completion of Y . Such a spectral sequence was first developed classically for mod 2 homology to study the 2-primary stable homotopy groups of spheres, i.e. $E = H\mathbf{Z}/2$, $X = \mathbf{S}^0 = Y$. The vast generality in which Adams spectral sequences have been shown to exist is a major reason why it is often easier to conduct computations stably within the homotopy category of spectra rather than the category of spaces. As we will be using the Adams spectral sequence to compute the 2-completed homotopy groups of Madsen Tillmann spectra, our computations will involve the mod 2 Adams spectral sequence

$$E_{s,t}^2 = \text{Ext}_{\mathcal{A}^*}(\mathbf{Z}/2, H_*(Y)) \Rightarrow \pi_{t-s}(Y)_p^\wedge$$

where $H_*(\mathbf{S}^0) = \tilde{H}_0(S^0) = \mathbf{Z}/2$, hence this is the version whose construction we will briefly develop. All of the material presented here is review which can be found in [Rav86]. Another excellent expository sources for the construction and use of the Adams spectral sequence is of course [Ada74].

We will also make the choice to dualize and instead work with the cohomological version of the mod 2 Adams spectral sequence, namely,

$$(8.1) \quad E_2^{s,t} = \text{Ext}_{\mathcal{A}}(H^*(X), \mathbf{Z}/2) \Rightarrow \pi_{t-s}(X)_p^\wedge.$$

Throughout, we assume X is a connective spectrum such that $H^*(X)$ is of finite type, for example, X is the suspension spectrum of a CW complex with finitely many cells in each dimension.

Definition 8.0.2. The *Hurewicz map* for X is defined as the composite

$$h_X: X \xrightarrow{\simeq} S^0 \wedge X \xrightarrow{\eta \wedge 1_X} H\mathbf{Z}/2 \wedge X.$$

The induced map $\pi_*(h_X)$ is called the *Hurewicz homomorphism*.

The motivation behind the Adams spectral sequence is as follows. The (stable) Hurewicz theorem along with the universal coefficient theorem states that the Hurewicz homomorphism is an isomorphism

$$(h_X)_n: \pi_n(X) \xrightarrow{\cong} H_n(X) \cong H^n(X)^\vee = \text{Hom}_{\mathbf{Z}/2}(H^n(X), \mathbf{Z}/2),$$

for n the smallest non-vanishing degree of $\pi_*(X)$.

In an ideal world, we would ask that the Hurewicz homomorphism be an isomorphism in all degrees, i.e., we would have an isomorphism of graded abelian groups

$$(h_X)_*: \pi_*(X) \cong [\mathbf{S}^0, X]_* \xrightarrow{\cong} \text{Hom}_{\mathcal{A}}(H^*(X), \mathbf{Z}/2)_* .$$

$$[f] \longmapsto H^*(f)$$

One need only notice how spectacularly this fails for the sphere spectrum (whose cohomology is concentrated in a single degree, but whose (2-completed) homotopy groups are vastly more complicated) to understand this is a unrealistic hope.

Rather than ask for such a restrictive property to be satisfied by all spectra, a more modest request would be to have a sufficiently large class $\mathcal{K}$ of such spectra, so that we could resolve (the cohomology ring of) any reasonably nice spectrum X by elements K_n of $\mathcal{K}$. That is, suppose we had a map of spectra $f_0: X \rightarrow K_0$ which induced an epimorphism on singular cohomology. We would thus have a short exact sequence

$$0 \rightarrow H^*(\Sigma \text{hofib}(f)) \rightarrow H^*(K) \rightarrow H^*(X) \rightarrow 0$$

of $\mathcal{A}$ -modules. The homotopy type of the fiber $X_1 := \text{hofib}(f)$ captures all of the homotopical information of X which can be detected by singular cohomology, in the following sense. Since $K \in \mathcal{K}$ is a class of spectra whose Hurewicz maps induce isomorphisms, $\pi_*(K_0)$ is completely understood, and the long exact sequence of

homotopy groups associated to the fiber sequence $X_0 \rightarrow X \xrightarrow{f} K_0$ implies the task of computing the homotopy groups of X reduces to the task of computing the homotopy groups of X_0 . We can subsequently iterate this process, obtaining a filtration $X_\bullet \rightarrow X$ of X which induces a filtration at the level of homotopy groups, and whose cofibers $K_n := \text{hocfib}(X_{n+1} \rightarrow X_n)$ induce a resolution of $H^*(X)$ by free $\mathcal{A}$ -modules.

The Adams spectral sequence is what refines this rough idea into a valid strategy for computing homotopy groups of spectra.

Definition 8.0.3. An *Adams resolution* (X_n, g_n) of X is a diagram in the stable homotopy category of the form

$$\begin{array}{ccccc} \dots & \xrightarrow{g_2} & X_2 & \xrightarrow{g_1} & X_1 & \xrightarrow{g_0} & X_0 = X \\ & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 \\ & & K_2 & & K_1 & & K_0 \end{array}$$

such that for each $n \geq 0$,

- $X_{n+1} \xrightarrow{g_n} X_n \xrightarrow{f_n} K_n$ is a fiber sequence;
- $H^*(g_n) = 0$, or equivalently $H^*(f_n)$ is surjective;
- $H^*(K_n)$ is a projective $\mathcal{A}$ -module;
- the Hurewicz homomorphism of K_n is an isomorphism.

Remark 8.0.4. Note the final condition in Definition 8.0.3 implies the seemingly stronger statement that, for each n , the Hurewicz map h_{K_n} induces a natural isomorphism

$$(h_{K_n})_* : [-, K_n]_* \xrightarrow{\cong} [-, H\mathbf{Z}/2 \wedge K_n]_*.$$

This follows from the Yoneda lemma, along with the fact the sphere spectrum $\mathbf{S}^0$ is a compact generator (in the sense of [5.6 Foundations Stable Homotopy]) of the stable homotopy category, and hence $\pi_*(-)$ detects isomorphisms.

We now define the spectra which make up the class $\mathcal{K}$, out of which we will build our Adams resolutions.

Definition 8.0.5. Let V be a graded $\mathbf{Z}/2$ vector space. A spectrum K is a *generalized (mod 2) Eilenberg-MacLane spectrum of type V* if it has the homotopy type of the Eilenberg-MacLane spectra HV of V

Note the Eilenberg-MacLane spectrum construction $A \mapsto HA$ defines an additive functor H from the category of (graded) abelian groups to the stable homotopy category. Thus if $K \simeq HV$ is a generalized Eilenberg-MacLane spectrum, such that the n^{th} -graded component V_n of V has $\mathbf{Z}/2$ -dimension k_n , we have

$$K \simeq H\left(\bigoplus_n (\mathbf{Z}/2)^{\oplus k_n}\right) \simeq \bigvee_n \Sigma^n (H\mathbf{Z}/2)^{\vee k_n}.$$

This implies generalized Eilenberg-MacLane spectra satisfy a number of nice properties, including those outlined in Definition 8.0.3.

Proposition 8.0.6. Let K be a generalized Eilenberg-MacLane spectrum of type V and X be an arbitrary spectrum.

1. $H^*(K)$ is finite type if and only if V is.
2. If V is of finite type, the Hurewicz homomorphism for K is an isomorphism.
3. A map $f: X \rightarrow K$ is equivalent to a collection of elements $\mathcal{U} \subset H^*(X)$

Proposition 8.0.7. The smash product $H\mathbf{Z}/2 \wedge X$ is an Eilenberg-MacLane spectrum of type $H^*(X)$, and the Hurewicz map h_X induces the $\mathcal{A}$ -module structure map on cohomology:

$$\mathcal{A} \otimes_{\mathbf{Z}/2} H^*(X) \cong H^*(H\mathbf{Z}/2 \wedge X) \xrightarrow{H^*(h_X)} H^*(X).$$

Thus, one can define an Adams resolution of X inductively by setting $K_n := H\mathbf{Z}/2 \wedge X_n$, $f_n := h_{X_n}$, and $X_n := \text{cofib}(f_n)$. This is known as the *standard Adams resolution* for X . Propositions 8.0.6 and 8.0.7 guarantee that such maps define an Adams resolution of X , hence we subsequently obtain our desired resolution of $H^*(X)$ by free $\mathcal{A}$ -modules:

$$\dots \rightarrow H^*(\Sigma^2 K_2) \rightarrow H^*(\Sigma K_1) \rightarrow H^*(K_0) \rightarrow H^*(X).$$

While applying singular cohomology to an Adams resolution gives us a relatively simple algebraic description relating the cohomology of the filtration $X_\bullet$ to the cohomology of X , the situation concerning homotopy groups is less straightforward.

Applying $\pi_*(-)$ to an Adams resolution for X yields a diagram of graded abelian groups

$$\begin{array}{ccccc} \dots & \xrightarrow{(g_2)_*} & X_2 & \xrightarrow{(g_1)_*} & \pi_*(X_1) & \xrightarrow{(g_0)_*} & \pi_*(X_0) \\ & & \downarrow (f_2)_* & & \downarrow (f_1)_* & & \downarrow (f_0)_* \\ & & \pi_*(K_2) & & \pi_*(K_1) & & \pi_*(K_0) \end{array}$$

such that each cofiber sequence $X_{n+1} \xrightarrow{g_n} X_n \xrightarrow{f_n} K_n \rightarrow \Sigma X_{n+1}$ induces a long exact sequence of homotopy groups

$$(8.8) \quad \dots \rightarrow \pi_*(X_{n+1}) \xrightarrow{(g_n)_*} \pi_*(X_n) \xrightarrow{(f_n)_*} \pi_*(K_n) \xrightarrow{\partial_{n,*}} \pi_{*-1}(X_{n+1}) \rightarrow \dots$$

This algebraic data forms an *exact couple*, i.e., a (non-commuting!) diagram of bigraded abelian groups

$$(8.9) \quad \begin{array}{ccc} D_1^{*,*} & \xrightarrow{i_1} & D_1^{*,*} \\ & \swarrow k_1 & \searrow j_1 \\ & E_1^{*,*} & \end{array}$$

where $D_1^{s,t} := \pi_{t-s}(X_s)$, $E_1^{s,t} := \pi_{t-s}(K_s)$, and the maps are given by

$$i_1 = (g_s)_{t-s}: \pi_{t-s}(X_s) \longrightarrow \pi_{t-s}(X_{s-1})$$

$$j_1 = (f_s)_{t-s}: \pi_{t-s}(X_s) \longrightarrow \pi_{t-s}(K_s)$$

$$k_1 = \partial_{s,t}: \pi_{t-s}(X_s) \longrightarrow \pi_{t-s-1}(X_{s+1})$$

and hence are of bidegrees $(-1, -1)$, $(0, 0)$, and $(1, 0)$ respectively. The exactness of 8.8 implies the following relations for each pair of indices $s, t \geq 0$ ⁷:

$$(8.10) \quad \ker j_1^{s,t} = \text{im } k_1^{s+1,t+1}$$

$$(8.11) \quad \ker k_1^{s,t} = \text{im } j_1^{s,t}$$

$$(8.12) \quad \ker i_1^{s,t} = \text{im } k_1^{s-1,t+2}$$

We can define a differential of bidegree $(1, 0)$ on $E_1^{*,*}$ by $d_1 := j_1 k_1: E_1^{s,t} \rightarrow E_1^{s+1,t}$. The relation 8.11 implies $d_1^2 = 0$, and passing to the cohomology $E_2 := H^{*,*}(E_1, d_1)$ of the double cochain complex of the exact couple yields an associated *derived exact couple*

$$\begin{array}{ccc} D_2^{*,*} & \xrightarrow{i_2} & D_2^{*,*} \\ & \swarrow k_2 & \searrow j_2 \\ & E_2^{*,*} & \end{array}$$

with $D_2^{*,*} := \text{im}(i_1)$, i_2 induced by i_1 , $j_2(i_1(d)) := j_1(d)$, and k_2 induced by k_1 . One can verify via a diagram chase that this does in fact define an exact couple (in the sense that the above diagram satisfies relations 8.10, 8.11, 8.12). Thus we can repeat the above process to obtain yet another derived couple. Iterating this procedure yields a sequence of exact couples

$$\begin{array}{ccc} D_r^{*,*} & \xrightarrow{i_r} & D_r^{*,*} \\ & \swarrow k_r & \searrow j_r \\ & E_r^{*,*} & \end{array}$$

⁷As X was assumed to be connective, $\pi_t(X_s) = 0$ for $t < 0$.

with $D_r^{*,*} := \text{im}(i_{r-1})$ and $d_r := j_{r-1}k_{r-1}$ a map of bidegree $(r, r-1)$. This construction defines a spectral sequence $\{(E_r^{*,*}, d_r)\}$. For $s < r$, the differential $d_r: 0 = E_r^{s-r, t-r+1} \rightarrow E_r^{s,r}$ is zero and thus $E_{r+1}^{s,t} = \ker d_r$ is a subobject of $E_r^{s,t}$. This allows us to define the E_∞ -page of the spectral sequence by

$$E_\infty^{s,t} := \bigcap_{r>s} E_r^{s,t}.$$

The assumptions we have made on X (namely, connective and with cohomology of finite type) guarantees the spectral sequence converges to the above E_∞ -page. This bigraded abelian group is the associated graded object of a particular decreasing filtration of the (2-completed) homotopy groups of X , which we develop below.

Definition 8.0.13. The *(mod 2) Adams spectral sequence* for X is the spectral sequence $E_r^{*,*}(X)$ associated to the exact couple 8.9.

8.1. Identifying the E_∞ -page: The Adams filtration. Recall the definition of the Adams spectral sequence $E_r^{*,*}(X)$ for a spectrum X required an Adams resolution $X_\bullet \xrightarrow{g_\bullet} X$ of X . Fix such an Adams resolution of X once and for all, comforted by the fact that different choices of such resolutions lead to isomorphic spectral sequences.

An Adams resolution determines a filtration of the homotopy groups of X in the following way.

Definition 8.1.1. An element $[f] \in \pi_*(X)$ has *Adams filtration n* if it factors through the n^{th} level of the Adams resolution of X , i.e. if there exists a dashed lift making the following diagram commute:

$$\begin{array}{ccc} & & X_n \\ & \nearrow & \downarrow g_0 \cdots g_{n-1} \\ \mathbf{S}^n & \xrightarrow{f} & X = X_0. \end{array}$$

Equivalently, $[f]$ has Adams filtration n if it lies in the image of the composite homomorphism

$$\pi_*(X_n) \xrightarrow{(g_{n-1})_*} \pi_*(X_{n-1}) \xrightarrow{(g_{n-2})_*} \cdots \xrightarrow{(g_1)_*} \pi_*(X_1) \xrightarrow{(g_0)_*} \pi_*(X).$$

We denote by $F^n \pi_*(X)$ the subset $\pi_*(X)$ consisting of elements of Adams filtration n .

From the second definition, it is clear that if $[f]$ has Adams filtration n , then it has Adams filtration for all $k < n$, and furthermore that $F^n \pi_*(X)$ is a (graded) subgroup of $\pi_*(X)$, hence we have a well-defined decreasing filtration $F^\bullet \pi_*(X)$.

Let $\text{Gr}^\bullet \pi_*(X)$ be the associated graded with respect to the Adams filtration of $\pi_*(X)$, i.e., $\text{Gr}^n \pi_*(X)$ is the graded $\mathbf{Z}/2$ vector space

$$\text{Gr}^n \pi_*(X) := F^n \pi_*(X) / F^{n+1} \pi_*(X).$$

We now verify that under suitable conditions, the E_∞ -page of the Adams spectral sequence for X coincides with this associated graded.

Lemma 8.1.2. Suppose $\text{holim}_{\leftarrow} X_n \simeq *$. Then $E_\infty^{s,t}$ is the $(t-s)$ -degree summand of $\text{Gr}^s \pi_*(X)$, i.e.

$$E_\infty^{s,t} \cong F^s \pi_{t-s}(X) / F^{s+1} \pi_{t-s}(X).$$

Theorem 8.1.3 (Adams). *The Adams spectral sequence is independent of the choice of Adams resolution of X . Furthermore, for X of finite type and connective, the Adams spectral sequence for X converges strongly to $\pi_*(X) \otimes \hat{\mathbf{Z}}_2$, with respect to the Adams filtration as defined in 8.1.1.*

8.2. Multiplicativity of the Adams spectral sequence. Much of the power of the Adams spectral sequence $E_r^{*,*}(X)$ comes from naturality in regards to the spectrum X . The upside of this is perhaps most apparent when X is a module of a ring spectrum E . In such a case, the module structure map

$$(8.1) \quad \alpha: E \wedge X \rightarrow X$$

descends to a pairing at the level of Adams spectral sequences which satisfy a variety of nice structural properties. The situation is summarized by the following theorem (2.3.3 of Ravenel).

Theorem 8.2.2. *Let α be as in 8.1, and $2 \leq r \leq \infty$. We have an induced pairing of bigraded abelian groups*

$$\alpha_r: E_r^{*,*}(E) \otimes_{\mathbf{Z}/2} E_r^{*,*}(X) \rightarrow E_r^{*,*}(X)$$

(whose image we denote by $\cdot$) satisfying the following properties:

1. (Leibniz rule). *Given $a \in E_r^{s,t}(E)$ and $x \in E_r^{s',t'}(X)$, we have*

$$d_r(a \cdot x) = d_r(a) \cdot x + a \cdot d_r(x).$$

2. *For $r = \infty$, the pairing coincides (up to Adams filtration) with the map $\alpha_*: \pi_*(E) \otimes_{\mathbf{Z}/2} \pi_*(X) \rightarrow \pi_*(X)$ which takes a pair of homotopy classes $[f: \mathbf{S}^n \rightarrow E]$ and $[g: \mathbf{S}^m \rightarrow X]$ to the homotopy class of the composite of the external cup product*

$$\mathbf{S}^{n+m} \xrightarrow{\sim} \mathbf{S}^n \wedge \mathbf{S}^m \xrightarrow{f \wedge g} E \wedge X \xrightarrow{\alpha} X.$$

3. *For $r = 2$, the pairing coincides with the external cup product on Ext groups*

$$\text{Ext}_{\mathcal{A}}(H^*(E), \mathbf{Z}/2) \otimes_{\mathbf{Z}/2} \text{Ext}_{\mathcal{A}}(H^*(X), \mathbf{Z}/2) \longrightarrow \text{Ext}_{\mathcal{A}}(H^*(E) \otimes_{\mathbf{Z}/2} H^*(X), \mathbf{Z}/2)$$

which is induced by the map on inner-hom $\mathcal{A}$ -modules given by the composite of

$$\underline{\text{Hom}}_{\mathcal{A}}(M, N) \otimes_{\mathbf{Z}/2} \underline{\text{Hom}}_{\mathcal{A}}(M', N') \rightarrow \underline{\text{Hom}}_{\mathcal{A}}(M \otimes_{\mathbf{Z}/2} M', N \otimes_{\mathbf{Z}/2} N'),$$

together with the map $\underline{\text{Hom}}_{\mathcal{A}}(H^(E) \otimes_{\mathbf{Z}/2} H^*(X), \mathbf{Z}/2) \rightarrow \underline{\text{Hom}}_{\mathcal{A}}(H^*(X), \mathbf{Z}/2)$ given by pulling back along*

$$H^*(X) \xrightarrow{\alpha^*} H^*(E \wedge X) \xrightarrow[\cong]{\text{Kunneth}} H^*(E) \otimes_{\mathbf{Z}/2} H^*(X).$$

4. *For $X = E$, the (commutative) ring structure of E endows the pages $E_r^{*,*}(E)$ with the structure of a (commutative) differential graded algebra.*

Example 8.2.3 ($\mathbf{S}^0$ -algebras). The most useful instance of the above theorem is also the most ubiquitous, namely the case where $E = \mathbf{S}^0$. Indeed, as $\mathbf{S}^0$ is the unit in the stable homotopy category with respect to the monoidal smash product, the coherence isomorphism $\mathbf{S}^0 \wedge X \xrightarrow{\sim} X$ witnesses every spectrum canonically as a module over the sphere spectrum. Thus by the above, for every spectrum X , the Adams spectral sequence $E_r^{*,*}(X)$ is a module over $E_r^{*,*}(\mathbf{S}^0)$, in an appropriate (graded) sense. Such extra structure plays a crucial role both in ruling out the existence of differentials, as well as solving extension problems to pass from the associated graded given by the E_∞ -page to the actual homotopy groups of the spectrum in question.

9. THE HOMOTOPY GROUPS OF $MTO(2)$

The spectrum $MTO(1)$ is 2-local. It follows that after inverting the prime 2, the cofiber sequence

$$MTO(2) \xrightarrow{\alpha_{O(2)}} \Sigma_+^\infty BO(2) \rightarrow MTO(1) \rightarrow \Sigma MTO(2)$$

implies that the MTW map $\alpha_{O(2)}$ yields a homotopy equivalence $MTO(2)[\frac{1}{2}] \simeq \Sigma_+^\infty BO(2)[\frac{1}{2}]$. Defining the group homomorphism

$$i: O(2) \rightarrow SO(3)$$

$$A \mapsto \det(A) \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$$

we have $\text{hofib}(Bi) \simeq SO(3)/O(2) \simeq \mathbf{R}P^6$. As $\mathbf{R}P^{2n}[\frac{1}{2}]$ is contractible, it follows that Bi is a homotopy equivalence after localizing away from 2.

In addition, the double covering $\mathbf{Z}/2 \rightarrow SU(2) \rightarrow SO(3)$ induces a fibration $f: BSU(2) \rightarrow BSO(3)$ whose homotopy fiber $B\mathbf{Z}/2 \simeq \mathbf{R}P^\infty$ is 2-local, hence f induces a homology isomorphism away from 2. In particular, we have $BO(2)[\frac{1}{2}] \simeq \mathbf{H}P^\infty[\frac{1}{2}]$. As localization at Moore spectra commutes with taking infinite suspensions, we get a homotopy equivalence

$$MTO(2) \left[\frac{1}{2} \right] \simeq \Sigma_+^\infty \mathbf{H}P \left[\frac{1}{2} \right],$$

i.e., the homotopy type of $MTO(2)$ away from 2 is well understood.

Warning 9.0.1. Throughout, we implicitly assume all spectra in sight are completed at the prime 2 unless stated otherwise.

9.1. A warm-up computation: the homotopy groups of $MTO(1)$. The task of computing the homotopy groups of the Madsen-Tillmann spectrum $MTO(1)$ is made significantly easier by the following result, found in chapter 1.5 of [Rav86].

Lemma 9.1.1. *The cofiber of the composite*

$$\omega_O : MTO(1) \xrightarrow{\alpha_{O(2)}} \Sigma^\infty \mathbf{R}P^\infty \vee \mathbf{S}^0 \xrightarrow{\pi_2} \mathbf{S}^0$$

coincides with the cofiber of the Kahn-Priddy map $\mathbf{R}^\infty \mathbf{R}P^\infty$. In particular, the composite induces an epimorphism on homotopy groups in positive degree.

We are able to determine differentials in the Adams spectral sequence for $MTO(1)$ up to topological degree 20.

Proposition 9.1.2.

1. *There is no d_2 -differential supported on $E_2^{1,6}$ as $\pi_3(\mathbf{S}^0) \cong \mathbf{Z}/8$, hence $E_2^{3,7}$ cannot be the target of a nonzero differential.*
2. *There are no generators in degree $t - s = 8$ which are the target of a differential. The exact sequence*

$$\pi_7(MTO(1)) \longrightarrow (\mathbf{Z}/16)^2 \oplus \mathbf{Z}/2 \longrightarrow \mathbf{Z}/16 \longrightarrow \mathbf{Z}/2 \longrightarrow (\mathbf{Z}/2)^2 \longrightarrow \mathbf{Z}/2 \longrightarrow 0$$

together with the Adams chart for $\Sigma MTO(1)$ imply $\pi_7(MTO(1)) \cong \mathbf{Z}/16 \oplus \mathbf{Z}/2$, from which the above result follows.

3. *There is no nontrivial differential supported on $t - s = 11$. The long exact sequence of homotopy groups associated to the cofiber sequence $MTO(1) \rightarrow \mathbf{R}P_+^\infty \rightarrow \mathbf{S}^0$ yields the exact sequence*

$$0 \longrightarrow \mathbf{Z}/8 \longrightarrow \mathbf{Z}/8 \longrightarrow \mathbf{Z}/8 \longrightarrow \pi_{10}(MTO(1)) \longrightarrow (\mathbf{Z}/2)^2 \oplus \mathbf{Z}/8 \longrightarrow \mathbf{Z}/2 .$$

This implies $\mathbf{Z}/8 \leq \pi_{10}(MTO(1))$. There is no hidden h_0 -extension between $E_2^{3,14}$ and $E_2^{6,17}$ as $E_2^{6,17}$ admits a nontrivial h_1 -action, while $h_1 h_0 = 0$ in $E_2^{s,t}(\mathbf{S}^0)$. From this it follows that $\pi_{10}(MTO(1)) \cong \mathbf{Z}/8 \oplus \mathbf{Z}/2$, completing the proof.

4. *We have the exact sequence*

$$\pi_{27}(MTO(1)) \rightarrow (\mathbf{Z}/8)^2 \rightarrow \mathbf{Z}/8 .$$

The kernel of the latter map contains $\mathbf{Z}/8$ as a subgroup, hence $\pi_{27}(MTO(1))$ contains a subgroup of order 8 or larger which implies $\pi_{27}(MTO(1))$ via the Adams chart for $\Sigma MTO(1)$. In particular, it follows that the above exact sequence is the unique split exact extension in $\text{Ext}^1(\mathbf{Z}/8, \mathbf{Z}/8)$. This implies no nontrivial differentials have source or target in the column $t - s = 28$.

5. *We have the exact sequence*

$$0 \rightarrow \pi_{28}(MTO(1)) \rightarrow (\mathbf{Z}/2)^2 \rightarrow \mathbf{Z}/2 \xrightarrow{0} \mathbf{Z}/8 .$$

In particular, we conclude $\pi_{28}(MTO(1)) \cong \mathbf{Z}/2$, hence there are two nontrivial differentials whose target lies on $t - s = 28$. It follows that $E_2^{9,38}$ and $E_2^{10,39}$ are both hit by d_2 differentials.

9.2. Splitting Madsen-Tillmann spectra.

The main insight we will leverage to compute the homotopy groups of $MTO(2)$ is the splitting result 1.0.2 of Kashiwabara and Zare.

The summand $D(2)$ is part of a sequence of spectra $D(n)$, $n \geq 0$, introduced by Mitchell and Priddy in [MP83]. We briefly motivate and recount their construction. For the moment, let us work integrally (i.e. within the usual stable homotopy category).

Recall given a space X , the n^{th} symmetric product $\text{SP}^n(X)$ is defined as the orbit space of the n -fold product X^n by the action of the symmetric group Σ_n given by permuting the factors:

$$\text{SP}^n(X) := X^n / \Sigma_n .$$

We generalize this construction to spectra as follows. Given a space X , we have a natural transformation

$$\begin{aligned} S^k \times \text{SP}^n(X) &\longrightarrow \text{SP}^n(S^k \times X) \\ (s, [x_1, \dots, x_n]) &\longmapsto [(s, x_1), \dots, (s, x_n)] . \end{aligned}$$

This mapping respects basepoints in each variable, and hence descends to map $\Sigma^k \mathrm{SP}^n(X) \rightarrow \mathrm{SP}^n(\Sigma^k X)$, also natural in X . Given a (sequential) spectrum $E = \{(E_k, \sigma_k)\}$, we define the *symmetric product* $\mathrm{SP}^n(E)$ to be the (functorial) fibrant replacement of the prespectrum with k^{th} -space given by $\mathrm{SP}^n(E_k)$ and k^{th} attaching map given by the composite

$$\Sigma \mathrm{SP}^n(E_k) \rightarrow \mathrm{SP}^n(\Sigma E_k) \xrightarrow{\mathrm{SP}^n(\sigma_k)} \mathrm{SP}^n(E_{k+1}) .$$

This construction results in a well-defined endofunctor $\mathrm{SP}^n(-)$ on $\mathrm{Ho}(\mathbf{Sp})$ which commutes with homotopy colimits.

The inclusion into the first n -factors $X^n \rightarrow X^{n+1}$ induces comparison maps $i_n: \mathrm{SP}^n(X) \rightarrow \mathrm{SP}^{n+1}(X)$ at the level of spaces (and consequently at the level of spectra as well). We define the *infinite symmetric product* $\mathrm{SP}^\infty(E)$ of a spectrum E as the homotopy colimit

$$\mathrm{SP}^\infty(E) := \operatorname{colim}_{n \rightarrow \infty} \mathrm{SP}^n(E)$$

with structure maps given by the i_n 's. In the case of the sphere spectrum, this construction takes on a familiar form.

Theorem 9.2.1. $\mathrm{SP}^\infty(\mathbf{S}^0) \simeq H\mathbf{Z}$.

Proof. As $\mathrm{SP}^\infty(\mathbf{S}^0)$ is a colimit of connective spectra, it is itself connective. Thus by the stable Hurewicz theorem, the Hurewicz map $h: \mathrm{SP}^\infty(\mathbf{S}^0) \rightarrow H\mathbf{Z} \wedge \mathrm{SP}^\infty(\mathbf{S}^0)$ induces an isomorphism on π_0 . It suffices to prove

$$\pi_k(\mathrm{SP}^\infty(\mathbf{S}^0)) \cong \begin{cases} \mathbf{Z}, & k = 0 \\ 0, & o/w \end{cases}$$

as by the universal coefficient theorem, we have cohomology isomorphisms $H^0(\mathrm{SP}^\infty; \mathbf{Z}) \cong \mathbf{Z} \cong H^0(H\mathbf{Z}; \mathbf{Z})$. It follows that the map $\iota: \mathrm{SP}^\infty(\mathbf{S}^0) \rightarrow H\mathbf{Z}$ corresponding to the unit in $H^0(\mathrm{SP}^\infty(\mathbf{S}^0); \mathbf{Z}) \cong [\mathrm{SP}^\infty(\mathbf{S}^0), H\mathbf{Z}]$ will induce an equivalence on $H^0(-; \mathbf{Z})$. The Hurewicz homomorphism implies such an ι induces an equivalence on π_0 , and hence yields the desired homotopy equivalence. We have

$$\begin{aligned} \pi_k(\mathrm{SP}^\infty(\mathbf{S}^0)) &= \pi_k(\operatorname{hocolim}_{n \rightarrow \infty} \mathrm{SP}^n(\mathbf{S}^0)) \\ &\cong \operatorname{colim}_{n \rightarrow \infty} \pi_k(\mathrm{SP}^n(\mathbf{S}^0)) \\ &\cong \operatorname{colim}_{n \rightarrow \infty} \pi_k(\operatorname{hocolim}_{l \rightarrow \infty} \Sigma^{-l} \mathrm{SP}^n(S^l)) \\ &\cong \operatorname{colim}_{n \rightarrow \infty} \operatorname{colim}_{l \rightarrow \infty} \pi_{k+l}(\mathrm{SP}^n(S^l)) \\ &\cong \operatorname{colim}_{l \rightarrow \infty} \operatorname{colim}_{n \rightarrow \infty} \pi_{k+l}(\mathrm{SP}^n(S^l)) \\ &\cong \operatorname{colim}_{l \rightarrow \infty} \pi_{k+l}(\operatorname{colim}_{n \rightarrow \infty} \mathrm{SP}^n(S^l)) \\ &\cong \operatorname{colim}_{l \rightarrow \infty} \pi_{k+l}(\mathrm{SP}^\infty(S^l)) \end{aligned}$$

The Dold-Thom theorem finishes the proof. $\square$

We now define a number of modules over the Steenrod algebra $\mathcal{A}$, each one realized as the cohomology of a spectrum constructed from constructions involving the symmetric products of the sphere spectrum. We first need a preliminary definition characterizing Steenrod squares which our modules of interest.

Definition 9.2.2. Let $I = (i_1, \dots, i_n)$ be a sequence of nonnegative integers $i_k \geq 0$. The *length* $\ell(I)$ of I is the number of nonzero terms in the sequence.

Definition 9.2.3. Fix $n \geq 0$.

1. We let D_n denote the vector subspace of $\mathcal{A}$ spanned by monomials Sq^I of length $\ell(I) \leq n$.
2. We let G_n denote the vector subspace of $\mathcal{A}$ spanned by admissible monomials Sq^I of length $\ell(I) \geq n$.

It is immediately clear from the definition along with the Adem relations that G_n is an $\mathcal{A}$ -module. We need some further justification that the $\mathbf{Z}/2$ -vector space D_n is indeed an $\mathcal{A}$ -module, i.e., closed under left multiplication by Steenrod squares.

Firstly, note that given a Steenrod square Sq^I of length $l(I) = n$, applying the Adem relations expresses Sq^I as unique linear combination of admissible monomials, each of length *at most* n , hence D_n has a basis consisting of *admissible* monomials of length at most n .

Given $n \geq 0$, the module G_n is in fact a left ideal of $\mathcal{A}$. Furthermore, one readily sees that the quotient $\mathcal{A}$ -module $\mathcal{A}/G_{n+1}$ has a vector space basis given by admissible monomials of length at most n , i.e., $\mathcal{A}/G_{n+1}$ realizes D_n as an $\mathcal{A}$ -module.

We also have inclusions of left ideals $G_{n+1} \subset G_n$ for every $n \geq 0$. The corresponding subquotient

$$M_n := G_n/G_{n+1}$$

of $\mathcal{A}$ canonically embeds into $\mathcal{A}$ as the linear subspace spanned by admissible sequences Sq^I of length $l(I) = n$.

Nakaoka determined that the filtration

$$\mathbf{S}^0 \rightarrow \mathrm{SP}^2(\mathbf{S}^0) \rightarrow \mathrm{SP}^{2^2}(\mathbf{S}^0) \rightarrow \dots \rightarrow \mathrm{SP}^{2^n}(\mathbf{S}^0) \rightarrow \dots \rightarrow H\mathbf{Z}$$

with the connecting map given by the composites of the maps $i_n : \mathrm{SP}^n(\mathbf{S}^0) \rightarrow \mathrm{SP}^{n+1}(\mathbf{S}^0)$, realizes the *length filtration* on $H^*(H\mathbf{Z}) \cong \mathcal{A}/\mathcal{A}Sq^1$, in the following sense. First, we need some notation.

Notation 9.2.4. Given an $\mathcal{A}$ -module M , denote by $\widetilde{M}$ the corresponding $\mathcal{A}/Sq^1\mathcal{A}$ -module induced by the extension of scalars along the ring homomorphism $\mathcal{A} \rightarrow \mathcal{A}/Sq^1\mathcal{A}$.

Theorem 9.2.5 ([Nak58]). *Fix $n \geq 0$.*

1. *If $n+1 \neq 2^k$ for some $k \geq 0$, the map $i_n : \mathrm{SP}^n(\mathbf{S}^0) \rightarrow \mathrm{SP}^{n+1}(\mathbf{S}^0)$ is a homotopy equivalence.*
2. *We have an isomorphism of $\mathcal{A}$ -modules*

$$H^*(\mathrm{SP}^{2^n}(\mathbf{S}^0)) \cong \widetilde{D}_n$$

and furthermore, the fiber sequence associated to the induced map $\mathrm{SP}^n(\mathbf{S}^0) \rightarrow H\mathbf{Z}$ realizes the short exact sequence of $\mathcal{A}$ -modules

$$0 \rightarrow \widetilde{G}_{n+1} \rightarrow \mathcal{A}/Sq^1\mathcal{A} \rightarrow \widetilde{D}_n \rightarrow 0.$$

We now go back to assuming all spectra are completed at the prime 2. Given a space X , the l -fold diagonal $\Delta_l : X \rightarrow X^l$ induces by naturality a map of symmetric products

$$\Delta'_l : \mathrm{SP}^n(X) \rightarrow \mathrm{SP}^n(X^l).$$

The natural map $(X^l)^n \simeq X^{ln} \rightarrow \mathrm{SP}^{ln}(X)$ factors through the Σ_n -orbits of the source, hence we can consider the composite map

$$\Delta_l : \mathrm{SP}^n(X) \xrightarrow{\Delta'_l} \mathrm{SP}^n(X^l) \rightarrow \mathrm{SP}^{ln}(X).$$

This space-level construction induces a map of prespectra, which by the functoriality of fibrant replacement, yields a natural transformation $\mathrm{SP}^n(-) \rightarrow \mathrm{SP}^{ln}(-)$ of endofunctors on $\mathrm{Ho}(\mathbf{Sp})$.

Given $n \geq 1$, we define the spectrum $D(n)$ to be the cofiber of the map

$$\Delta_2 : \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \rightarrow \mathrm{SP}^{2^n}(\mathbf{S}^0).$$

By convention $\mathrm{SP}^{2^{-1}}(\mathbf{S}^0)$ has the homotopy type of a point, so that $D(0) = \mathrm{SP}^1(\mathbf{S}^0) = \mathbf{S}^0$.

Let $p_n : \mathrm{SP}^{2^n}(\mathbf{S}^0) \rightarrow \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0)$ denote the image of the degree p map on $\mathbf{S}^0$. Since $\mathrm{SP}^{2^n}(-)$ preserves cofiber sequences, we have $\mathrm{cofib}(p_n) \simeq \mathrm{SP}^{2^n}(M(\mathbf{Z}/p))$. Mitchell and Priddy show Δ_p factors as the composite

$$\begin{array}{ccc} \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) & \xrightarrow{\Delta_p} & \mathrm{SP}^{2^n}(\mathbf{S}^0) \\ & \searrow p_{n-1} & \nearrow \\ & \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) & \end{array}$$

where the latter map is the composite of the filtration maps $i_k : \mathrm{SP}^k(\mathbf{S}^0) \rightarrow \mathrm{SP}^{k+1}(\mathbf{S}^0)$ for $2^{n-1} \leq k \leq 2^n - 1$ (and whose notation we suppress from commutative diagrams for the remainder of this section).

Thus, we get a cofiber sequence of filtered spectra

$$\begin{array}{ccccccc}
 * & \longrightarrow & \mathbf{S}^0 & \longrightarrow & \mathrm{SP}^2(\mathbf{S}^0) & \longrightarrow & \dots \longrightarrow H\mathbf{Z} \\
 \Delta_p \downarrow & & \Delta_p \downarrow & & \Delta_p \downarrow & & \downarrow p \\
 \mathbf{S}^0 & \longrightarrow & \mathrm{SP}^2(\mathbf{S}^0) & \longrightarrow & \mathrm{SP}^{2^2}(\mathbf{S}^0) & \longrightarrow & \dots \longrightarrow H\mathbf{Z} \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 \mathbf{S}^0 & \longrightarrow & D(1) & \longrightarrow & D(2) & \longrightarrow & \dots \longrightarrow H\mathbf{Z}/p
 \end{array}$$

Mitchell and Priddy go on to prove a realization theorem for the above filtration of $H\mathbf{Z}/p$ analogous to Nakaoka's theorem for $H\mathbf{Z}$.

Theorem 9.2.6. *We have an isomorphism of $\mathcal{A}$ -modules*

$$H^*(D(n); \mathbf{Z}/2) \cong \mathcal{A}/G_{n+1}$$

and furthermore, the map $D(n) \rightarrow H\mathbf{Z}/2$ realizes the projection map $\mathcal{A} \rightarrow \mathcal{A}/G_{n+1}$ on $\mathbf{Z}/2$ -cohomology.

Finally, given $n \geq 0$, we define the spectrum $L(n)$ as the n -fold desuspension of the cofiber of the map $j_n: \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) \rightarrow \mathrm{SP}^{2^n}(\mathbf{S}^0)$ corresponding to the inclusion $(\mathbf{S}^0)^{2^{n-1}} \rightarrow (\mathbf{S}^0)^{2^n}$ of the first 2^{n-1} -factors. The situation is summarized by the following commutative diagram of spectra, where the rows and the columns are cofiber sequences:

$$\begin{array}{ccccc}
 \mathrm{SP}^{2^{n-2}}(\mathbf{S}^0) & \xrightarrow{\Delta_p} & \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) & \longrightarrow & D(n-1) \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathrm{SP}^{2^{n-1}}(\mathbf{S}^0) & \xrightarrow{\Delta_p} & \mathrm{SP}^{2^n}(\mathbf{S}^0) & \longrightarrow & D(n) \\
 \downarrow & & \downarrow & & \downarrow \\
 \Sigma^{n-1}L(n-1) & \longrightarrow & \Sigma^n L(n) & \longrightarrow & \Sigma^n M(n)
 \end{array}$$

For $n = 1$, Mitchell and Priddy showed that the above cofiber sequences take on a familiar form.

Proposition 9.2.8 (Theorem 3.2 of [MP84]). *We have homotopy equivalences*

$$D(1) \simeq \Sigma MTO(1), \quad L(1) \simeq \Sigma^\infty \mathbf{R}P^\infty, \quad M(2) \simeq \Sigma_+^\infty \mathbf{R}P^\infty.$$

9.3. Computing the E_2 -pages and analyzing the Adams charts. We are now ready to compute the homotopy groups of $D(2)$ and $\Sigma^\infty BSO(3)$. The Adams E_2 -pages depicted in figures 1, 2.1, 2.2, and 3.1 below are computed via minimal resolutions generated using Bruner's Ext software [Bru22]. We begin by determining differentials in the Adams spectral sequence for $D(2)$.

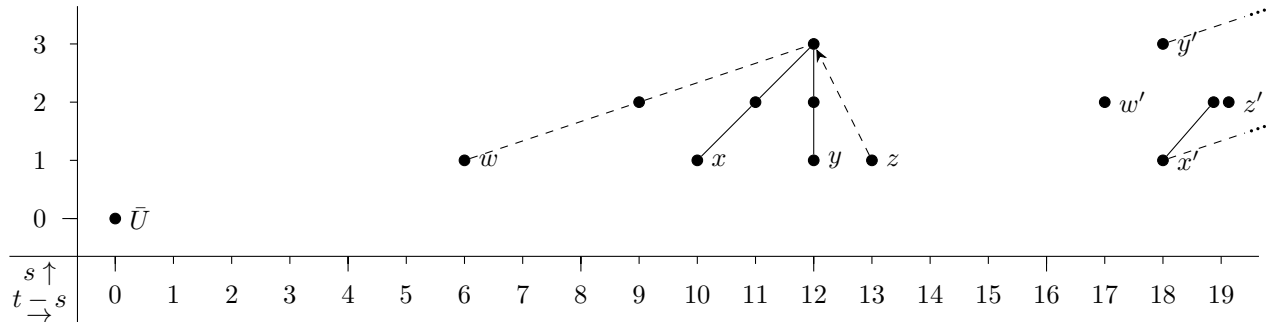


FIGURE 1. $\mathrm{Ext}_{\mathcal{A}}(D(2), \mathbf{Z}/2) = \mathrm{Ext}_{\mathcal{A}}(D_2, \mathbf{Z}/2)$

In topological degrees $t - s \leq 18$, there is only room for a single differential with nonzero target in the form of a d_2 -differential on $E_2^{14,1}$.

Proposition 9.3.1. *The d_2 -differential $d_2: E_2^{1,14}(D(2)) \rightarrow E_2^{3,15}(D(2))$ vanishes.*

Proof. The cofiber sequence $D(1) \rightarrow D(2) \rightarrow \Sigma^2 M(2)$ yields a long exact sequence of Ext groups relating the E_2 -pages of the respective Adams spectral sequences of $D(1)$, $D(2)$, and $\Sigma^2 M(2)$. We also have an associated long exact sequence of homotopy groups

$$\dots \rightarrow \pi_{13}(D(1)) \rightarrow \pi_{13}(D(2)) \rightarrow \pi_{13}(\Sigma^2 M(2)) \rightarrow \pi_{12}(D(1)) \rightarrow \dots$$

As $D(1) \simeq \Sigma MTO(1)$, we know the homotopy groups of $D(1)$ in the relevant ranges, namely we have the following exact of homotopy groups

$$0 \rightarrow \pi_{13}(D(2)) \rightarrow \pi_{13}(\Sigma^2 M(2)) \rightarrow \mathbf{Z}/8 \rightarrow \pi_{12}(D(2)) \rightarrow \pi_{12}(\Sigma^2 M(2)) \rightarrow \mathbf{Z}/2 \oplus \mathbf{Z}/8 \rightarrow \mathbf{Z}/2.$$

It suffices to conclude $\pi_{13}(\Sigma^2 M(2)) = \mathbf{Z}/2 \times \mathbf{Z}/8$. Looking at the Adams spectral sequence for $\Sigma^2 M(2)$, the splitting $M(2) \simeq L(2) \vee \mathbf{RP}^\infty$ induces a splitting of E_2 -pages

$$\mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(\Sigma^2 M(2)), \mathbf{Z}/2) \cong \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(\Sigma^2 L(2)), \mathbf{Z}/2) \oplus \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(\Sigma^2 \mathbf{RP}^\infty), \mathbf{Z}/2).$$

This splitting persists on every page E_r of the spectral sequence, such that for $r = \infty$ it corresponds (up to Adams filtration) to the splitting of homotopy groups. The Adams spectral sequence and stable homotopy groups of $\mathbf{RP}^\infty$ are well-known (page 82, table IV of [Whi70]). In particular, $E_r^{*,*}(\mathbf{RP}^\infty)$ admits no nontrivial differentials in topological degrees $t - s \leq 13$, and from this we can conclude that there is no differential supported in topological degree 13 in the Adams spectral sequence for $\Sigma^2 M(2)$. This completes the proof. $\square$

First, we claim the cofiber sequence $MTSO(2) \xrightarrow{\beta} \Sigma MTSO(3) \xrightarrow{-\Sigma\alpha} \Sigma BSO(3)_+ \xrightarrow{\partial} \Sigma MTSO(2)$ yields the following short exact sequence on cohomology groups:

$$(9.2) \quad 0 \rightarrow H^*(\Sigma BSO(3)_+) \rightarrow H^*(\Sigma MTSO(3)) \rightarrow H^*(MTSO(2)) \rightarrow 0.$$

Indeed, we have the following exact sequence on cohomology groups associated to the cofiber sequence:

$$0 \rightarrow H^0(\Sigma MTSO(3)) \rightarrow H^0(MTSO(2)) \xrightarrow{\partial^*} H^1(\Sigma BSO(3)_+) \rightarrow H^1(\Sigma MTSO(3)) \rightarrow 0.$$

Since $H^0(\Sigma MTSO(3)) \cong H^{-1}(MTSO(3)) \cong \mathbf{Z}/2\{w_2 \overline{U}'\}$ and $H^0(MTSO(2)) \cong \mathbf{Z}/2\{w_2 \overline{U}\}$, where $\overline{U}$ and $\overline{U}'$ are the respective Thom classes for the virtual vector bundles $-\gamma_2^{SO}$ and $-\gamma_3^{SO}$, it follows that $\partial^*(\overline{U}) = 0$. As ∂^* is an $\mathcal{A}$ -module homomorphism, $Sq^2(\overline{U}) = w_2 \overline{U}$, and the fact that $H^*(MTSO(2))$ is the free $\mathbf{Z}/2[w_2]$ -module on a single generator $\overline{U}$, it follows that ∂^* is the zero map on cohomology, yielding the short exact sequence in 9.2. Thus by the geometric boundary theorem (2.3.4 in [Rav86]), we have an algebraic boundary map

$$\delta: \mathrm{Ext}_{\mathcal{A}}^{s,t}(\Sigma BSO(3)_+, \mathbf{Z}/2) \rightarrow \mathrm{Ext}_{\mathcal{A}}^{s+1,t}(MTSO(2), \mathbf{Z}/2)$$

on E_2 -pages of the respective Adams spectral sequences which commutes with differentials. Using the notation

$$\begin{aligned} E_2^{s,t} &:= \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(\Sigma BSO(3)_+), \mathbf{Z}/2), \\ {}'E_2^{s,t} &:= \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(\Sigma MTSO(3)), \mathbf{Z}/2), \\ {}''E_2^{s,t} &:= \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(MTSO(2)), \mathbf{Z}/2), \end{aligned}$$

and $i := \omega_*$ and $j := (-\Sigma\beta)_*$ for the induced maps on Ext groups, the cofiber sequence above yields a long exact sequence of Ext groups

$$(9.3) \quad \dots \rightarrow {}''E_2^{s,t} \xrightarrow{i} {}'E_2^{s,t} \xrightarrow{j} E_2^{s,t} \xrightarrow{\delta} {}''E_2^{s+1,t} \rightarrow \dots$$

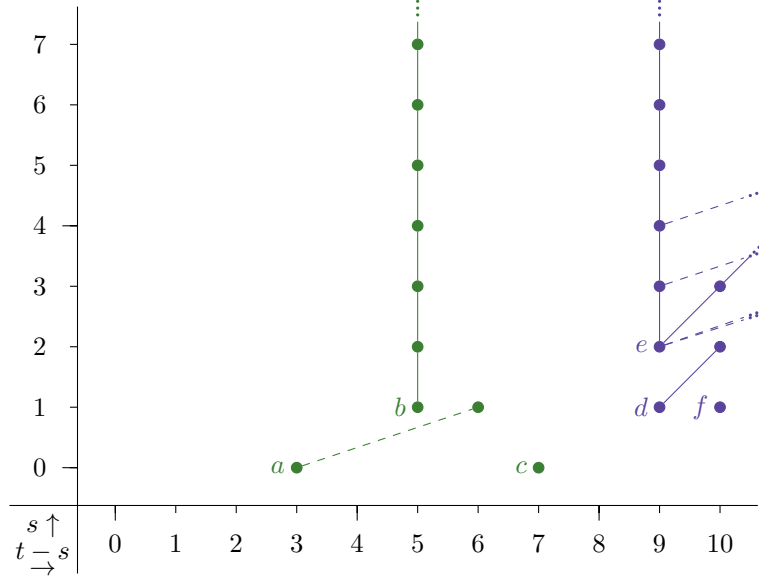
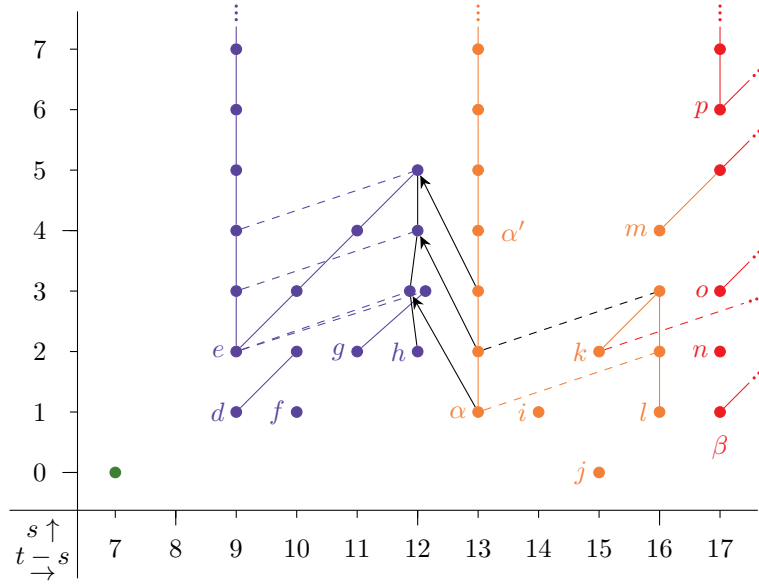
This will allow us to transfer differentials from $E_r^{*,*}(MTSO(2))$ to $E_r^{*,*}(BSO(3))$ under suitable conditions.

Our approach rely heavily on the work of Rognes in [Rog02] (pages 894-898), in which he computed the Adams spectral sequence for $MTSO(2)$ and determined all differentials up through degree 20.

The fact that differentials are linear with respect to the $E_2^{*,*}(\mathbf{S}^0)$ -module structure of $E_2^{*,*}$ precludes the existence of nontrivial differentials in topological degrees $0 \leq t - s \leq 11$ and $13 \leq t - s \leq 15$. The following proposition determines all nontrivial differentials in the Adams spectral sequence for $BSO(3)$ up to and including topological degree 16.

Proposition 9.3.4.

(1) *There exists a nontrivial d_2 -differential $d_2: E_2^{1,14} \rightarrow E_2^{3,15}$ with target h_0h . Consequently by h_0 -linearity, there are nontrivial d_2 -differentials supported on $E_2^{1,15}$ and $E_2^{2,16}$.*

FIGURE 2.1. $E_2 = \text{Ext}_{\mathcal{A}}(H^*(\Sigma BSO(3)), \mathbf{Z}/2)$ FIGURE 2.2. $E_2 = \text{Ext}_{\mathcal{A}}(H^*(\Sigma BSO(3)), \mathbf{Z}/2)$

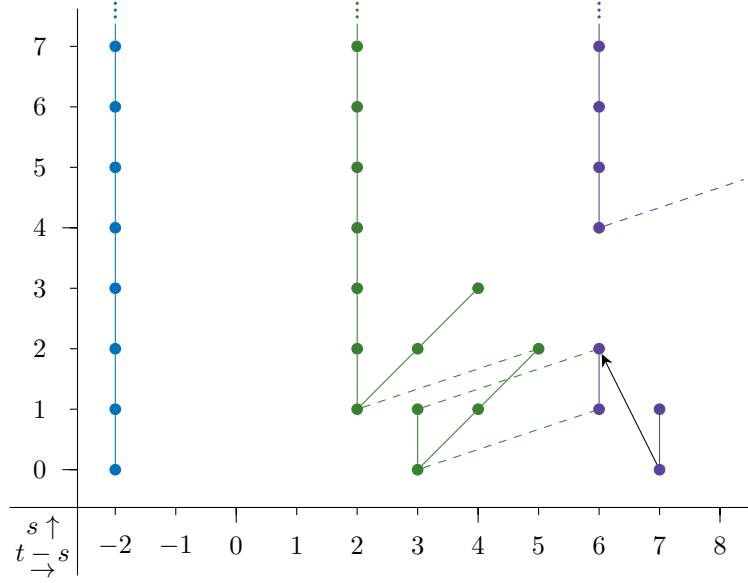
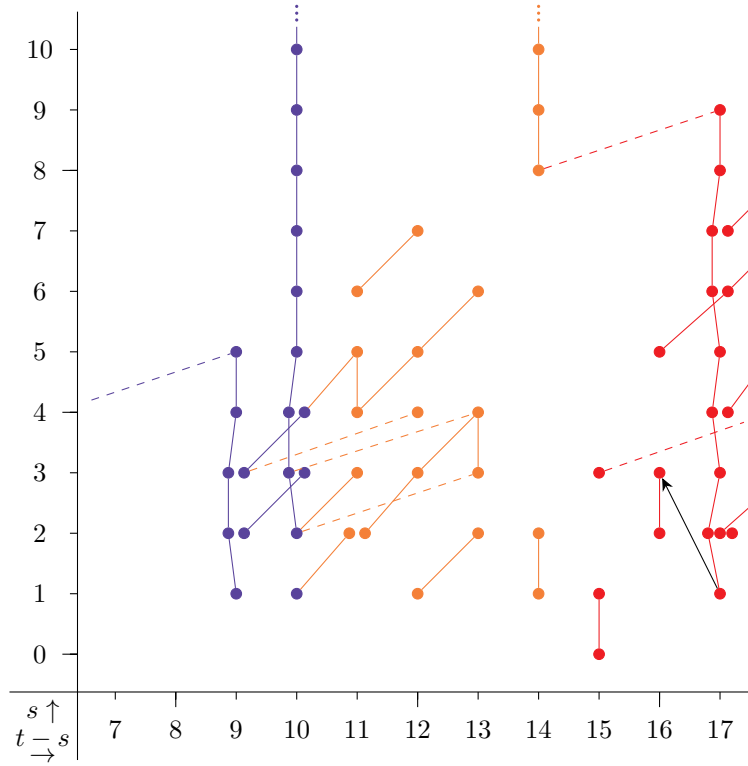
(2) There are no nontrivial differentials supported on $E_2^{1,18}$.

Proof.

(1) The map of spectral sequences $\delta: E_2^{s,t} \rightarrow {}''E_2^{s+1,t}$ is an isomorphism in bidegree $(1, 14)$. Indeed, the long exact sequence 9.3 of Ext-groups yields the fiber sequence

$$'E_2^{1,14} = 0 \rightarrow E_2^{1,14} \cong \mathbf{Z}/2\{\alpha\} \xrightarrow{\delta} {}''E_2^{2,14} \cong \mathbf{Z}/2\{h_0 h_3\}.$$

As the image of the generator of $E_2^{1,14} \cong \mathbf{Z}/2\{\alpha\}$ along δ supports a nonzero d_2 -differential on $MTSO(2)$, this implies α does as well. It remains to determine the image of this differential. We have $E_2^{2,14} \cong \mathbf{Z}/2\{h\}$ and ${}''E_2^{3,14} \cong \mathbf{Z}/2\{h_0 h_3\} \oplus \mathbf{Z}/2\{h_1^3\}$ using Rognes' notation. The map $\delta: E_2^{2,14} \rightarrow {}''E_2^{3,14}$ is an injection by the same argument above (the fiber is a subgroup of $'E_2^{2,14} = 0$). As $h_1 h = 0$, it follows that $i(h)$ must have trivial h_1 -action, and so we conclude $i(h) = h_0 h_3$. Then h_0 -linearity of i implies $i(h_0 h) = h_0^2 h_3 = {}''d_2(i(\alpha))$, hence $d_2(\alpha) = h_0 h$ as required.

FIGURE 3.1. $'E_2 = \text{Ext}_{\mathcal{A}}(H^*(\Sigma MTSO(3)), \mathbf{Z}/2)$ FIGURE 3.2. $'E_2 = \text{Ext}_{\mathcal{A}}(H^*(\Sigma MTSO(3)), \mathbf{Z}/2)$

(2) We claim the map $\delta: E_2^{s,t} \rightarrow ''E_2^{s+1,t}$ is an isomorphism in bidegrees (3,19) and (4,20) and injective in bidegree (1,18). Let α denote the generator of $E_2^{1,18}$.

Since the d_2 -differential on $''E_2^{2,18}$ is zero, by injectivity of δ in degrees (1,17) and (3,18), it follows that $d_2: E_2^{1,18} \rightarrow E_2^{3,19}$ is zero as well.

We have $E_3^{4,20} \cong E_2^{4,20}$ and $''E_3^{5,20} \cong ''E_2^{5,20}$, hence δ descends to an isomorphism between these terms on the E_3 -pages. Furthermore, we have $E_3^{1,18} \cong E_2^{1,18}$, $''E_3^{2,18} \cong ''E_2^{2,18}$, and so $\delta: E_3^{1,18} \rightarrow E_3^{2,18}$ remains an injection. Since $0 = d_3(\delta(\alpha)) = \delta(d_3(\alpha))$, we conclude $d_3(\alpha) = 0$. This completes the proof.

□

To determine the homotopy groups of $D(2)$ and $BSO(3)$ from the Adams E_∞ -pages we determined in the range $t - s \leq 16$, we need to solve the corresponding extension problems to pass from the associated graded objects of the Adams filtration 8.1.1 to the actual homotopy groups. Again, here the $E_r^{*,*}(\mathbf{S}^0)$ -module structure of Adams spectral sequences comes in handy. Because h_0 -action on E_2 -pages converges to multiplication by 2 on homotopy groups induced by the natural $\mathbf{S}^0$ -module structure of spectra, we can rule out the existence of certain hidden extensions. For example, if there existed a hidden h_0 extension $e = h_0 d$ in degree $t - s = 9$ of the Adams spectral sequence for $\Sigma BSO(3)$, since $h_1 e \neq 0$, this would imply at the level of homotopy groups the relation $2\eta d \neq 0$ in $\pi_9 BSO(3)$, but the Hopf map $\eta \in \pi_1(\mathbf{S}^0)$ has order 2, hence such an extension cannot exist.

With this in mind, the following theorem summarizes our computations of the 2-primary homotopy groups of $MTO(2)$ through the 16^{th} stem. Extension problems which we could not resolve are distinguished by the symbol ‘ $\ltimes$ ’. Relative to the 2-complete splitting $MTO(2) \simeq \Sigma^{-2}D(2) \vee \Sigma^\infty BSO(3) \vee \mathbf{S}^0$, generators of $\pi_*(MTO(2))$ which lie in $\pi_*(\Sigma^{-2}D(2))$, $\pi_*(\mathbf{S}^0)$, or $\pi_*(\Sigma^\infty BSO(3))$, are denoted blue, green or orange respectively. The notation for the generators for $\pi_*(\mathbf{S}^0)$ is taken from page 82, table IV of [Whi70].

Theorem 9.3.5. *The 2-primary homotopy groups $\pi_*(MTO(2))$ in dimensions $* \leq 15$ are given as follows:*

k	$\pi_k MTO(2)_{(2)}$
-2	$\mathbf{Z}/2\{\textcolor{blue}{U}\}$
-1	0
0	$\mathbf{Z}\{\textcolor{blue}{U}\}$
1	$\mathbf{Z}/2\{\eta\}$
2	$\mathbf{Z}/2\{\textcolor{blue}{a}\} \oplus \mathbf{Z}/2\{\textcolor{green}{\eta}^2\}$
3	$\mathbf{Z}/8\{\nu\}$
4	$\mathbf{Z}/2\{\textcolor{blue}{w}\} \oplus \mathbf{Z}\{\textcolor{blue}{b}\}$
5	$\mathbf{Z}/2\{\nu a\}$
6	$\mathbf{Z}/2\{\textcolor{blue}{c}\} \oplus \mathbf{Z}/2\{\nu^2\}$
7	$\mathbf{Z}/2\{\nu w\} \oplus \mathbf{Z}/16\{\sigma\}$
8	$\mathbf{Z}/2\{\textcolor{blue}{x}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{d}\} \oplus \mathbf{Z}\{\textcolor{blue}{e}\} \oplus \mathbf{Z}/2\{\eta\sigma\} \oplus \mathbf{Z}/2\{\epsilon\}$
9	$\mathbf{Z}/2\{\eta\textcolor{blue}{x}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{f}\} \ltimes \mathbf{Z}/2\{\eta\textcolor{blue}{d}\} \oplus \mathbf{Z}/2\{\eta\textcolor{blue}{e}\} \oplus \mathbf{Z}/2\{\nu^3\} \oplus \mathbf{Z}/2\{\eta\epsilon\} \oplus \mathbf{Z}/2\{\mu\}$
10	$\mathbf{Z}/8\{\textcolor{blue}{y}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{g}\} \ltimes \mathbf{Z}/2\{\eta^2\textcolor{blue}{e}\} \oplus \mathbf{Z}/16\{\nu\mu\}$
11	$\mathbf{Z}/2\{\textcolor{blue}{z}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{h}\} \ltimes \mathbf{Z}/2\{\eta\textcolor{blue}{g}\} \oplus \mathbf{Z}/8\{\zeta\}$
12	$\mathbf{Z}\{\alpha'\}$
13	$\mathbf{Z}/2\{\textcolor{blue}{i}\}$
14	$\mathbf{Z}/2\{\textcolor{blue}{j}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{k}\} \oplus \mathbf{Z}/16\{\sigma^2\} \oplus \mathbf{Z}/16\{\kappa\}$
15	$\mathbf{Z}/2\{\textcolor{blue}{w}'\} \oplus \mathbf{Z}/8\{\textcolor{blue}{l}\} \oplus \mathbf{Z}/2\{\textcolor{blue}{m}\} \oplus \mathbf{Z}/32\{\rho\} \oplus \mathbf{Z}/2\{\eta\kappa\}$

9.4. Looking ahead: the homology of the infinite non-orientable mapping class group. In Theorem C of [MP84], Mitchell and Priddy obtain a 2-complete splitting of $\Sigma_+^\infty BO(2)$:

$$\Sigma_+^\infty BO(2) \simeq \Sigma_+^\infty BSO(3) \vee M(2).$$

As finite products and coproducts in a triangulated category coincide, passing to infinite loop spaces (and noting that $\pi_0(M(2)) = 0$), this gives a homotopy equivalence

$$Q_0(BO(2)_+) \simeq Q_0(BSO(3)_+) \times \Omega^\infty M(2).$$

Lemma 4.10 of [CLM76] defines the $\mathbf{Z}/2$ -homology of the (basepoint component) of the free infinite loop space $Q_0 X$ functorially in terms of $H_*(X; \mathbf{Z}/2)$. Thus the above equivalence, combined with the Künneth isomorphism, should in allow us to determine the rank of the homology groups $H_n(\Omega^\infty M(2))$. From this, we should in principal be able to use the fibration sequence

$$0 \rightarrow \Omega_0^{\infty+1} MTO(1) \rightarrow \Omega_0^{\infty+2}(D(2)) \rightarrow \Omega_0^\infty(M(2)) \rightarrow 0$$

associated to the cofiber sequence in 9.7 to compute the rank of the homology groups of $\Omega_0^\infty(D(2))$. The 2-complete splitting $MTO(2) \simeq \Sigma_+^\infty BO(2)_+ \vee \Sigma^{-2}D(2)$ descends to the following homotopy equivalence of infinite loop spaces:

$$\Omega_0^\infty MTO(2) \simeq Q_0(BO(2)) \times \Omega_0^{\infty+2}(D(2)), .$$

The homotopy equivalence $B\mathcal{N}^+ \simeq \Omega_0^\infty MTO(2)$ implies the above outlines a potential method for computing the rank of the $\mathbf{Z}/2$ -homology groups of $B\mathcal{N}^+$ (and hence $B\mathcal{N}$). Since away from the prime 2, we have $B\mathcal{N}^+[\frac{1}{2}] \simeq Q_0(\mathbf{H}P^\infty)$, this is the only obstruction to determining the integral homology of the infinite non-orientable mapping class group (up to potential non-trivial extensions of the 2-primary part).

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